Differentiable transmon design with a tensor-native finite-element electromagnetics solver: gradient-based optimization of charging energy, cross-validated against Palace

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Abstract

Superconducting-qubit electromagnetic design is an iterative guess-and-check loop: the workflow's production field solvers expose no derivatives of design figures-of-merit, so state-of-practice optimizers iterate time-consuming electromagnetic simulations and inject parameter updates from separate analytic physics models (Eriksson et al., 2025). We demonstrate the missing capability — *solver-derived* design gradients from a 3D finite-element electromagnetics solve — and use it for gradient-based optimization of a transmon's charging energy E_C. GEODE-FEM is a Rust solver built on the Burn tensor framework; a discrete-adjoint layer bridges the one stage its autodiff tape cannot reach (the sparse factorization), yielding a validated 2×2 sensitivity matrix — material ($\partial/\partial\varepsilon$) and geometry ($\partial/\partial X$) derivatives on both the electrostatic and the driven $H(\mathrm{curl})$ operators — each finite-difference-validated with mutation tests (relative errors 10^{-9}–10^{-5}). Because the capacitance $C = \phi^{\mathsf{T}} K \phi$ is variationally stationary, its shape derivative collapses to a Hellmann–Feynman-like explicit term; composing it with the capacitance-to-E_C chain, Newton iteration drives a capacitor geometry to a 0.2156 GHz target E_C, confirmed by an independent fresh solve to 1.4×10^{-15}, and on the real 133k-tetrahedron device mesh $\partial C_\Sigma/\partial\theta$ is FD-validated at 1.15×10^{-4} with a genuine two-step convergence to a within-budget target. Harmonic mesh motion extends the safe deformation budget by 8.90×, but reaches only 128.793 fF toward the 89.9 fF anchor; the falling sensitivity implies a conditional extrapolated shortfall of at least 6.25× the available budget. A three-parameter, junction-pinned optimizer with per-step volume-stiffened re-morphing then drives the coarse-mesh C_Σ to 89.9 fF in 18 steps. That is a methods result, not a device design: one uniform refinement lowers C_Σ of the unmodified device by 26.3%, so those coarse results do not establish a design accuracy. A refined P2 extension reaches 89.9005 fF in three gradient steps; a held-out local refinement gives 89.6628 fF and independent Palace gives 89.9003 fF. These checks support local refinement stability, not a certified continuum error bound. Eigenmode/EPR derivatives remain roadmap. The correctness credential: from the identical input mesh, GEODE-FEM reproduces Palace's six eigenmodes above 4.5 GHz to $\leq 0.033\%$ (worst case 0.032%), and the transmon qubit mode at 3.45 GHz to 5×10^{-5} %. Earlier versions of this paper called that mode spurious and the qubit mode absent; a Palace run with its search target below the mode corrects both statements. The contribution is a capability, not a speed race — on raw performance the measured record shows no corner where GEODE-FEM beats Palace at scale, and we report it honestly. Every result traces to a committed artifact; statements that rest on a scratch probe, on a reading of Palace's source, or on inference are labelled as such.

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1 Introduction

The hard problem in superconducting-qubit electromagnetic design is the *inverse map*: given target Hamiltonian parameters — a charging energy E_C, an anharmonicity α, coupling capacitances — produce a physical layout. Today that map is walked by hand: parameterize a geometry, run a 3D electromagnetic simulation, extract the parameters, compare, adjust, repeat. The state-of-practice automated workflow makes the loop's structure explicit: QDesignOptimizer (Eriksson et al., 2025) iterates time-consuming electromagnetic simulations of HFSS-class solvers and guides the parameter updates with *separate*, user-defined analytic physics models — precisely because the field solver at the center of the loop produces observables but no derivatives of those observables with respect to the design. The gradient the optimizer actually wants, $\partial(\text{figure of merit})/\partial(\text{geometry, materials})$ through the field solve itself, is structurally unavailable from the production solvers — commercial (HFSS, COMSOL) and open-source (Palace (Palace developers, AWS Center for Quantum Computing, 2023)) alike: their assembly and factorization pipelines are hand-written kernels with no derivative path, and none exposes an adjoint of its solve.

The surrounding literature brackets this gap without closing it. At the *lumped-circuit* level, differentiation is a solved problem: SQcircuit (Rajabzadeh et al., 2023) analyzes arbitrary superconducting circuits, and its gradient-based follow-up (Rajabzadeh et al., 2024) computes eigenvalue and eigenvector gradients of the circuit Hamiltonian via PyTorch autodiff, optimizing qubit designs in circuit-parameter space. But the map from geometry and materials *to* those lumped parameters still runs through non-differentiable 3D electromagnetic simulation. At the *field* level, differentiable electromagnetics is owned by photonics: adjoint and autodiff methods on FDTD and integral-equation solvers are a mature inverse-design industry (Hammond et al., 2022; Hughes et al., 2019; Mahlau et al., 2026; Molesky et al., 2018; Ponomareva et al., 2025; Schubert et al., 2022), and differentiable FEM at scale has been proven outside electromagnetics by JAX-FEM in solid mechanics (Xue et al., 2023). What none of these supplies is the regime superconducting-qubit (and, more broadly, RF) design lives in: *frequency-domain finite elements on body-fitted tetrahedra with anisotropic materials* — distributed, field-level EM gradients from the same FEM solve that produces the design observables. That wedge is, to our knowledge, uncontested; this paper occupies it.

GEODE-FEM ("geode-fem") is positioned to close the gap by construction rather than retrofit. It is a full-wave 3D $H(\text{curl})$/Nédélec (plus scalar electrostatic) finite-element solver written in Rust on the Burn tensor framework (Tracel AI, 2024): element assembly is batched $[n_{\text{elem}}, 6, 6]$ tensor algebra on an autodiff-capable substrate, so the derivative of assembly with respect to materials and node coordinates is available where a classical solver has only hand-written kernels. One stage breaks the tape: the sparse direct factorization (faer (Kazdadi, 2026)) at the heart of every solve. A discrete-adjoint layer closes exactly that gap — one extra (transpose-)solve, reusing the forward factorization, restores the full gradient (Section 8). The resulting capability is general — shape and material sensitivities for RF, photonic, and electrostatic device design — and the transmon is our demonstration vehicle because its unmet need is the clearest documented in the literature, not because the method is qubit-specific.

A design-gradient claim from a new solver is only as credible as the solver's forward answers, so the paper carries its correctness credential up front: on a real transmon-plus-readout-resonator geometry meshed once and given as the same input file to both codes, geode-fem reproduces Palace — the MFEM-based (Anderson et al., 2021; Andrej et al., 2024) reference solver for superconducting-qubit design — to within 0.033% across the six eigenmodes Palace returns above 4.5 GHz (worst case 0.032%), and to 5×10^{-5} % on the transmon qubit mode at 3.45 GHz, with analytic scaling tripwires guarding against coincidental agreement (Section 5). Palace refines the mesh around the metal sheets internally, so the two discrete problems are not identical (Section 4.1). This parity is *a*

credential, not the contribution — matching Palace is the price of entry — and it is, incidentally, the first independent third-party validation of the Palace/DeviceLayout.jl transmon stack, which has no publication of its own.

Our contributions are:

- **A validated 2×2 discrete-adjoint sensitivity matrix for FEM electromagnetics.** Material ($\partial/\partial\varepsilon$) and geometry ($\partial/\partial X$, exact $\partial K/\partial X$ via forward-mode duals through the element kernels) derivatives, on both the scalar electrostatic and the complex driven $H(\mathrm{curl})$ Nédélec operators — each one forward plus one adjoint solve sharing a single factorization, each validated against full-pipeline central finite differences with mutation tests that reject sign, conjugation, and dropped-term errors (Section 8).

- **A differentiable capacitance$\to E_C$ chain with a variational-stationarity simplification.** Because $C = \phi^{\mathsf{T}} K \phi$ is stationary in the solved potential, the implicit term vanishes and the shape derivative is a pure explicit-geometry (Hellmann–Feynman-like) term; validated against the analytic parallel-plate law and central finite differences (Section 8.2).

- **Gradient-based optimization demonstrations, honest tiers labeled.** An end-to-end Newton loop that hits a 0.2156 GHz target E_C and is confirmed by an independent fresh solve at 1.4×10^{-15} (the parametrization is affine, so one-step convergence proves the loop, not a hard optimization); a real-device demonstration on the committed 133k-tet mesh with the gradient FD-validated at 1.15×10^{-4} and a genuine two-step convergence; and harmonic mesh motion that widens the safe budget 8.90× but still stalls at 128.793 fF toward the 89.9 fF anchor; and a three-parameter junction-pinned optimizer with per-step re-morphing that reaches 89.9 fF on the coarse mesh (Section 9). The last is a methods result: the device-mesh C_Σ is not mesh-converged, and the converged value is still being measured.

- **The correctness credential.** Same-input-mesh, same-physics, same-junction-model cross-validation against Palace at $\leq 0.033\%$ across the six modes above 4.5 GHz and 5×10^{-5} % on the qubit mode, with oracle-first methodology and honest-physics reporting — including the correction of two claims made in earlier versions, that the qubit mode was absent and that the 3.45 GHz mode was spurious (Sections 5 and 6) — doubling as the first independent validation of the Palace/DeviceLayout.jl stack.

- **Mode-verified performance context.** Same-box CPU measurements disclose mode selection, field-output and error-estimator costs; they replace the earlier unsupported one-core-versus-eight-rank efficiency claim. CPU-f64 AMS reduces driven-solve time relative to Jacobi, but its iteration and setup costs still grow with refinement (Section 11). Eigenmode/EPR derivatives remain roadmap (Section 10).

Section 2 positions the work; Section 3 presents the substrate and where its tape breaks; Sections 4–6 establish the benchmark problem and the correctness credential; Sections 7–9 are the contribution — the forward LOM chain, the adjoint layer, and the optimization demonstrations; Section 10 is the eigenmode roadmap; Sections 11–13 report performance context, reproducibility, and discussion.

2 Related Work

Superconducting-qubit design optimization. The design loop this paper attacks is documented across the qubit-design software stack. QDesignOptimizer (Eriksson et al., 2025) is the state

of practice: physics-guided multi-parameter optimization that wraps non-differentiable EM simulation (HFSS with pyEPR/Qiskit-Metal-class post-processing) and steers it with analytic models, because no gradient is available from the solver. The SQcircuit line (Rajabzadeh et al., 2023, 2024) differentiates at the lumped-circuit level — including eigenvalue/eigenvector gradients of the circuit Hamiltonian via PyTorch — and optimizes in circuit-parameter space; the geometry-to-parameters map remains simulation-bound. The SQuADDS database (Shanto et al., 2024) attacks the same loop by lookup: pre-simulated, experimentally-validated design points. Qiskit Metal's quasi-lumped methods (Minev et al., 2021b) and the EPR/black-box quantization physics (Koch et al., 2007; Minev et al., 2021a; Nigg et al., 2012) frame the parameter-extraction layer this paper's LOM chain implements. None of these obtains gradients from the field solve itself; that is the gap Section 8 closes.

Differentiable electromagnetics — owned by photonics. Adjoint/autodiff EM is a mature inverse-design discipline in nanophotonics (Molesky et al., 2018), expressed through structured grids and integral methods: autograd FDFD/FDTD in ceviche (Hughes et al., 2019), foundry-constrained JAX inverse design (Schubert et al., 2022), the Meep adjoint pipeline (Hammond et al., 2022), the JAX-native FDTDX framework (Mahlau et al., 2026), GPU-accelerated autodiff integral-method scattering in TorchGDM (Ponomareva et al., 2025), and differentiable RCWA (Kim et al., 2024). These codes accept rectilinear grids or Green's-function discretizations; the superconducting-qubit regime — body-fitted tetrahedra, anisotropic sapphire, $H(\mathrm{curl})$ conformity, frequency domain — is exactly where that literature stops. Differentiable FEM has been proven outside EM: JAX-FEM (Xue et al., 2023) (GPU-accelerated, autodiff-native continuum mechanics) and PyTorch-FEA (Liang et al., 2023) in biomechanics; concurrent work on general differentiable Galerkin assembly (TensorGalerkin (Wen et al., 2026), with the torch-sla adjoint sparse-solver library (Chi and Wen, 2026)) independently confirms the substrate-level thesis while reporting nodal magnetostatics rather than full-wave $H(\mathrm{curl})$ EM. To our knowledge, no prior work delivers field-level design gradients from a frequency-domain 3D FEM electromagnetics solve of the class RF/superconducting design uses.

FEM kernels by code generation, and tensor compilers. Generating FEM kernels from high-level descriptions is a two-decade tradition — FFC (Kirby and Logg, 2006), UFL (Alnæs et al., 2014), Firedrake (Rathgeber et al., 2017), TSFC (Homolya et al., 2018) — whose modern GPU expression is libCEED (Brown et al., 2021) and MFEM's partial-assembly path (Andrej et al., 2024). Palace itself runs on that domain-specific stack. geode-fem's substrate bet is categorically different: outsource kernels, scheduling, and backend portability to a *general-purpose ML tensor stack* (the Halide/TVM compiler lineage (Chen et al., 2018; Ragan-Kelley et al., 2013), reaching scientific computing via ML frameworks), and phrase assembly and matrix-free application as batched tensor algebra. For this paper the decisive property of that choice is not portability but *differentiability*: an autodiff-capable substrate makes the assembly derivative available by construction, which is what the adjoint layer of Section 8 composes with. Adjacent genera — simulation DSLs (Holl et al., 2020; Hu et al., 2019), neural-ansatz eigensolvers (Jiang et al., 2024), hand-written CUDA FEM (Meng et al., 2014) — trade away either the general-purpose stack or the discretization itself.

Validation and cross-solver benchmarking. Code-to-code comparison on precisely specified problems is an established genre in computational electromagnetics (the TEAM workshop tradition since 1986[1]; nano-optics solver comparisons (Hoffmann et al., 2009)); in V&V vocabulary (Oberkampf

[1]The TEAM founding documents are workshop reports without resolvable DOIs; we describe the tradition in prose.

and Roy, 2010; Roache, 1997) our credential is verification-side, code-to-code evidence. On the qubit-EM axis the closest works are SQDMetal (Sommers et al., 2025) (Palace cross-checked against COMSOL/HFSS at the 0.02–0.13% level, independently constructed models) and the Palace-centered workflow of Ye et al. (2025) (validated against cryogenic measurement at ∼0.3%). Both wrap and trust Palace; neither is a same-mesh comparison against an independently *implemented* solver. Our protocol removes mesh generation and model entry as confounders — identical MSH fixture, identical lumped-shunt junction, first-order Nédélec on both sides — so residuals reflect formulation, solver and Palace's internal refinement of the metal sheets (Section 4.1), not independently built models. Mature open EM tools normally carry a methods paper (GetDP (Dular et al., 1998), FEniCS/DOLFIN (Logg and Wells, 2010), Gmsh (Geuzaine and Remacle, 2009), scikit-rf (Arsenovic et al., 2022), MFEM (Anderson et al., 2021)); Palace and DeviceLayout.jl are unpublished outliers (AWS Center for Quantum Computing, 2025; AWS Quantum Technologies Blog, 2023; Palace developers, AWS Center for Quantum Computing, 2023; Peairs, 2025; Peairs and Carson, 2025), which is the gap the credential narrows from the outside. geode-fem's numerics: first-order Nédélec elements (Nédélec, 1980), shift-invert Lanczos (Lehoucq et al., 1998) over a real symmetric pencil factorized with faer (Kazdadi, 2026); Palace: SLEPc-class Krylov (Hernandez et al., 2005) with divergence-free projection. The timed Palace build resolves its default linear solver to SuperLU under GMRES; explicit AMS runs are reported separately (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009); the gauge-fixing literature behind Section 6 is the tree–cotree tradition (Albanese and Rubinacci, 1988; Manges and Cendes, 1995).

3 GEODE-FEM: A Tensor-Native Substrate, and Where the Tape Breaks

geode-fem is written in Rust on the Burn tensor framework (Tracel AI, 2024), whose cubecl JIT compiles tensor kernels for CPU, CUDA, and WebGPU backends from one source. Every throughput-critical FEM stage is phrased as the batched dense algebra the stack already optimizes.

Batched element-local assembly. For first-order Nédélec tetrahedra the element curl–curl and mass contributions are 6×6 matrices depending on per-element geometry factors and material tensors (including the rotated anisotropic sapphire permittivity of Section 4.1); the scalar electrostatic operator is the analogous P1 kernel. geode-fem computes them for all elements at once as batched $[n_{\text{elem}}, 6, 6]$ tensor operations — one kernel launch per assembly stage rather than an element loop — then either scatters into a global sparse matrix (the assembled f64 CPU path) or retains the batch for matrix-free application.

Matrix-free application and on-device Krylov. The matrix-free operator apply is a three-stage tensor pipeline — gather global DOFs to element-local vectors, batched matmul against the stored element operators, scatter-add back — and the driven frequency-sweep solve runs a conjugate orthogonal conjugate gradient (COCG) iteration entirely on the device with a constant number of scalar host–device synchronizations per iteration. The backend (ndarray on CPU, CUDA on GPU) is a type parameter, not a code path. Section 11 reports honestly how far below the GPU-amortization threshold the current benchmark sizes sit.

Honest constraints. Two boundaries are constraints, not footnotes. The CUDA backend (burn-cuda 0.21) is f32-only today — the underlying cubecl runtime currently disables f64[2] — so every GPU result is mixed-precision-qualified. And the eigensolve that produces the credential comparison

[2]Upstream constraint of the burn/cubecl stack at version 0.21 (Tracel AI, 2024); tracking-issue URL to be added at revision. TODO(operator).

is factorization-bound: its inner solves are faer sparse-LU factorizations (Kazdadi, 2026), a classical serial CPU component outside the tensor-compiler story.

Where the tape breaks — and why an adjoint layer, not autodiff-through-the-solver. The substrate choice has a consequence the rest of this paper builds on: Burn's automatic differentiation reaches everything *up to* the linear solve — material and node-coordinate dependence of the assembled operators is differentiable by construction — but the sparse factorization is an external, opaque kernel that breaks the tape. Differentiating *through* a factorization by unrolling is neither practical nor necessary: the classical discrete-adjoint identity turns the missing derivative into one additional linear solve against the *same* factorization (Section 8). The division of labor is deliberate — autodiff where the substrate provides it (element kernels, via forward-mode duals), adjoint algebra where it does not (the solve) — and it is what makes the gradients in this paper *solver-derived* in the sense the design loop of Section 1 lacks: the same assembly and the same factorization that produce the observable produce its gradient.

4 Benchmark Problem: Geometry, Mesh, and Junction Model

4.1 Geometry and mesh provenance

The device is the `SingleTransmon` example shipped with DeviceLayout.jl v1.15.0 (AWS Center for Quantum Computing, 2025): a transmon with readout resonator on a sapphire substrate, enclosed in a vacuum box. The sapphire is anisotropic, with permittivity tensor $\varepsilon = R\,\mathrm{diag}(9.3,\, 9.3,\, 11.5)\,R^{\mathsf{T}}$ rotated approximately 36.87° in-plane. The DeviceLayout.jl workflow emits seven named physical groups (metal, substrate, vacuum, junction lumped-element surface, two readout ports, exterior boundary) and drives Gmsh (Geuzaine and Remacle, 2009) to produce a tetrahedral mesh, exported as an MSH 4.1 fixture and pinned by SHA-256 (Section 12).

The mesh has 22,684 nodes and 133,314 tetrahedra, yielding 156,863 Nédélec edge unknowns, of which 133,108 interior degrees of freedom remain after eliminating perfect-electric-conductor (PEC) boundary edges. PEC is applied on all metal surfaces and the exterior box; the two readout ports are left open (lossless), so no resistive element enters and the eigenproblem pencil stays real symmetric. These counts are geode-fem's. Palace reads the same file but does not solve the same discrete problem: its run logs record 4,913 elements added by local bisection around under-resolved interior boundaries and 6,014 duplicated vertices where it cracks the metal sheets, and it assembles 183,891 first-order Nédélec unknowns against geode-fem's 156,863 (about 17% more). "Same mesh" in this paper therefore means the same *input* mesh. Earlier versions said the two solvers consumed an identical mesh and solved an identical discrete problem; that was wrong, and the timing comparisons of Section 11.1 carry the difference as a fairness caveat. This mesh is also coarse for the electrostatic quantities: one uniform refinement lowers C_Σ by 26.3% (Section 7). Figure 1 shows the geometry with physical groups color-coded. The eigenmode credential and initial electrostatic LOM and gradient demonstrations start from this committed mesh (plus a uniformly refined variant for the scale study of Section 11, and one uniform refinement for the C_Σ check of Section 7). The refined P2 experiment in Section 9.4 uses local refinement of the same polygonal geometry.

4.2 The junction model: a lumped reactive shunt

The Josephson junction is modeled identically in both solvers as a lumped reactive shunt — a parallel $L \parallel C$ element with $L = 14.860\,\mathrm{nH}$ and $C = 5.5\,\mathrm{fF}$ — attached to the four-triangle `lumped_element` surface group Γ. Following the uniform lumped-port construction, with port length ℓ and width w,

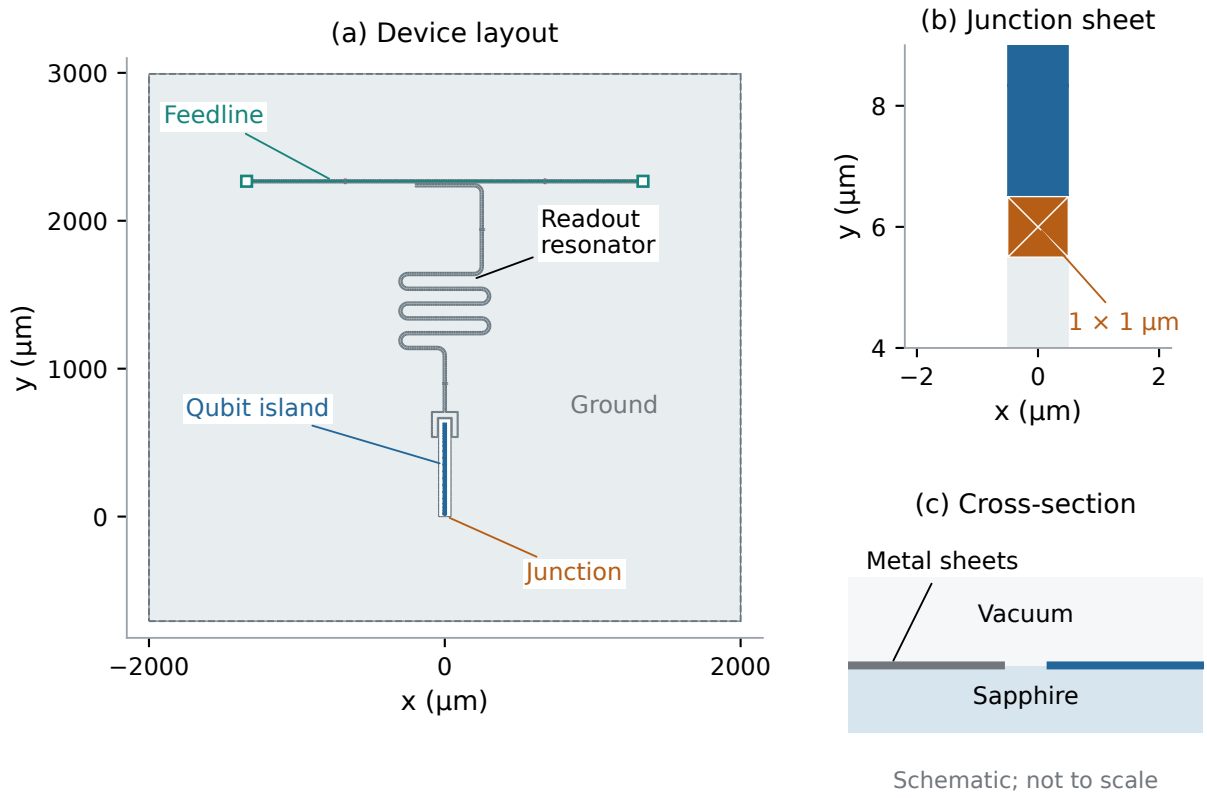


Figure 1: DeviceLayout SingleTransmon geometry. (a) Conductor outlines from the committed surface mesh: the feedline is teal, the qubit island blue, and the ground/resonator conductor gray; squares mark readout ports. (b) The actual $1 \times 1\,\mu$m junction sheet and its four triangles. (c) Material cross-section schematic, not to scale. The input mesh has 22,684 nodes and 133,314 tetrahedra. Palace internally refines near the metal sheets; the two solvers do not use identical discrete spaces.

the shunt contributes

$$K_{\text{port}} = \frac{\ell}{w\,L}\,S_\Gamma \qquad \text{and} \qquad M_{\text{port}} = \frac{C\,\ell}{w}\,S_\Gamma, \tag{1}$$

where S_Γ is the port-surface matrix assembled on Γ. The eigenproblem is then the real symmetric generalized pencil

$$(K + K_{\text{port}})\,x \;=\; \omega^2\,(M + M_{\text{port}})\,x. \tag{2}$$

In Palace, the same physics is entered as a reactive `LumpedPort` ($L \parallel C$, no resistance) on the same surface group; the readout ports are configured open, so Palace's pencil is likewise real. Palace's source confirms the correspondence: its `LumpedPort` adds the same scalar surface-mass terms with the same per-square scalings (`docs/research/transmon-lumped-port-formulation.md`, §1). This one-to-one correspondence is what makes the comparison same-physics as well as same-input-mesh. It also means that any property of the distributed surface term in Eq. 1 is shared by both solvers (Section 6.3).

5 The Correctness Credential: Same-Input-Mesh Cross-Validation Against Palace

A solver claiming design gradients must first prove its forward answers; this section is that proof. Palace's own agreement with commercial tools is established (Sommers et al., 2025), so reproducing Palace is the bar a new implementation must clear before anything it computes *about* its solutions (such as their derivatives) is worth discussing.

Protocol. The benchmark is oracle-first. The mesh is generated once, committed, pinned by SHA-256, and read by both solvers at matched (first) order with the identical junction model; residual disagreement is therefore attributable to formulation, implementation or Palace's internal refinement (Section 4.1), not to independent meshing. Two analytic tripwires guard against coincidental or self-consistent-but-wrong agreement. Both test the port term's L and C scalings. The first is the resonance of the port sheet's own inductance against its own capacitance, $f_{LC} = 1/(2\pi\sqrt{LC}) = 17.60\,\text{GHz}$, which anchors the 17.49 GHz mode. The second is a $\sqrt{L}$-scaling test: doubling L must move that mode by $1/\sqrt{2}$ — measured: 17.4901 → 12.37 GHz, committed ratio 0.7071 (computed from the unrounded values; Figure 3). Earlier versions called the 17.49 GHz mode "the junction mode" and treated these two checks as pinning the junction physics. Section 6.3 shows that this mode carries no port voltage, so the checks pin the port *operator*, not the qubit; the qubit mode has its own scaling check in Section 6.1. The suite also contains inverse tests that *must fail* (perturbed configurations required to disagree with the oracle — a pipeline that still passes under them is not measuring anything), and the Palace oracle is deterministic: a rerun from the committed configuration reproduced its committed `eig.csv` bit-for-bit.

Mode identification. Modes are identified, not eyeballed. Within geode-fem, the stiffness-participation ratio

$$p \;=\; \frac{x^{\mathsf{T}} K_{\text{port}}\, x}{x^{\mathsf{T}} (K + K_{\text{port}})\, x} \tag{3}$$

measures the share of a mode's stiffness energy stored in the port surface term, whatever its spatial pattern. It is $p = 1.000$ for the 17.49 GHz mode, $p = 0.9942$ for the 3.45 GHz mode, and $p = 0.000$ (at the artifact's three-decimal precision) for the resonator and box modes. Palace's port EPR is built from the port *voltage* functional alone, so it counts only the uniform-voltage part of that energy. The two diagnostics agree where the port field is a uniform voltage and disagree where it is not: on the qubit mode Palace reports 0.99424 against geode-fem's 0.9942, while on the 17.49 GHz mode Palace reports 2.49×10^{-8} against geode-fem's 1.000 (Section 6). In the original Palace run, which searched only above 4.5 GHz, all six port-EPR values are $\leq 5 \times 10^{-4}$; earlier versions read that as two diagnostics ranking the modes differently, when the cause was the qubit mode's absence from Palace's search window. The six-mode comparison of Table 1 is matched by frequency.

Result. Table 1: all six modes agree to $\leq 0.033\%$ (worst case 0.032%). The 17.49 GHz port-sheet mode agrees most closely, at 0.001%, and sits 0.6% below $f_{LC} = 17.60$ GHz in *both* solvers. Agreement that tight on a mode confined to the four port triangles is what an identical port operator would produce, and it is weaker evidence about the bulk discretizations than the cavity modes' ∼0.03%. The qubit mode, which Palace returns only when asked to search below it, agrees to $5 \times 10^{-5}\,\%$ (Section 6.1, Table 2). The agreement is between genuinely independent code paths — a batched-tensor assembly feeding a direct-factorization Lanczos in Rust, and a projected, preconditioned Krylov method in C++ — on discrete problems that differ by Palace's internal refinement (183,891

Table 1: Eigenmode agreement from the identical input mesh (22,684 nodes / 133,314 tets; geode-fem solves 156,863 Nédélec DOFs, 133,108 interior after PEC; Palace refines internally and solves 183,891), first order in both solvers, junction as a lumped reactive shunt ($L = 14.860\,\text{nH}$, $C = 5.5\,\text{fF}$). These are the six modes Palace returns when it searches above 4.5 GHz; the qubit mode below that target is in Table 2. geode-fem at commit `3174015`; Palace at commit `fba6a5b`. All values are quoted verbatim from the committed `benchmarks/transmon_eigen/results.toml`: geode-fem frequencies at the committed three decimals, Palace at full precision, and the committed per-mode `rel_err_pct`, computed against the rounded geode-fem values (precision disclosure in Section 12).

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
`resonator`	readout resonator	5.153	5.151335830348	0.032
`mode_2`	box/cavity	15.465	15.46052107794	0.029
`junction_lc`	port-sheet LC*	17.490	17.49010903536	0.001
`mode_4`	box/cavity	18.693	18.69165792915	0.007
`mode_5`	box/cavity	20.703	20.69755679425	0.026
`mode_6`	box/cavity	26.088	26.08089940472	0.027

*The artifact key `junction_lc` is kept verbatim. Palace's port EPR for this mode is 2.49×10^{-8}, and Section 6.3 identifies it as an artifact of the distributed port term.

against 156,863 unknowns). A common implementation error is therefore an unlikely explanation for the residual. A shared *modeling* property is not excluded by cross-comparison, and Section 6.3 reports one. All of this is same-input-mesh agreement at first order; none of these frequencies is a mesh-converged value.

6 Honest Physics: The Qubit Mode, Two Withdrawn Claims, and a Port-Sheet Artifact

Versions 1–6 of this paper made two claims in this section, and both were wrong. The first was that no qubit mode exists in this junction model by construction, and that Palace confirmed its absence. The second was that geode-fem's 3.4528 GHz mode is a spurious port artifact with no Palace counterpart. We withdraw both. The committed evidence identifies the 3.45 GHz mode as the transmon qubit mode, which Palace also returns, and it points to a different mode, at 17.49 GHz, as the artifact of the port model. This section separates what is proven by committed artifacts from what is derived, what a scratch probe measured, and what is only planned.

6.1 The 3.45 GHz mode is the transmon qubit mode

Why the earlier Palace run could not return it. The original Palace run asked for six modes with `Target` = 4.5 GHz and returned nothing below the 5.15 GHz resonator. Earlier versions read that as Palace hunting near 4.5 GHz and finding no qubit mode. That misreads `Target`. Palace's eigensolver accepts only eigenvalues at or above the target: its SLEPc configuration restricts the search region to that side, and its configuration schema defines `Target` as the frequency "above which to search" (`docs/research/transmon-lumped-port-formulation.md`, §2, read from the Palace source at the benchmarked changeset). The run's silence below 4.5 GHz was therefore not evidence of absence.

Proven: Palace returns the mode when asked. We re-ran Palace at the same changeset on the same input mesh, with the same materials, `LumpedPort`, order, eigensolver tolerances and linear solver, changing only `Target` (4.5 → 2.0 GHz) and the number of modes (6 → 8); a second cell also doubles L. A control cell with the original request reproduces the committed spectrum to

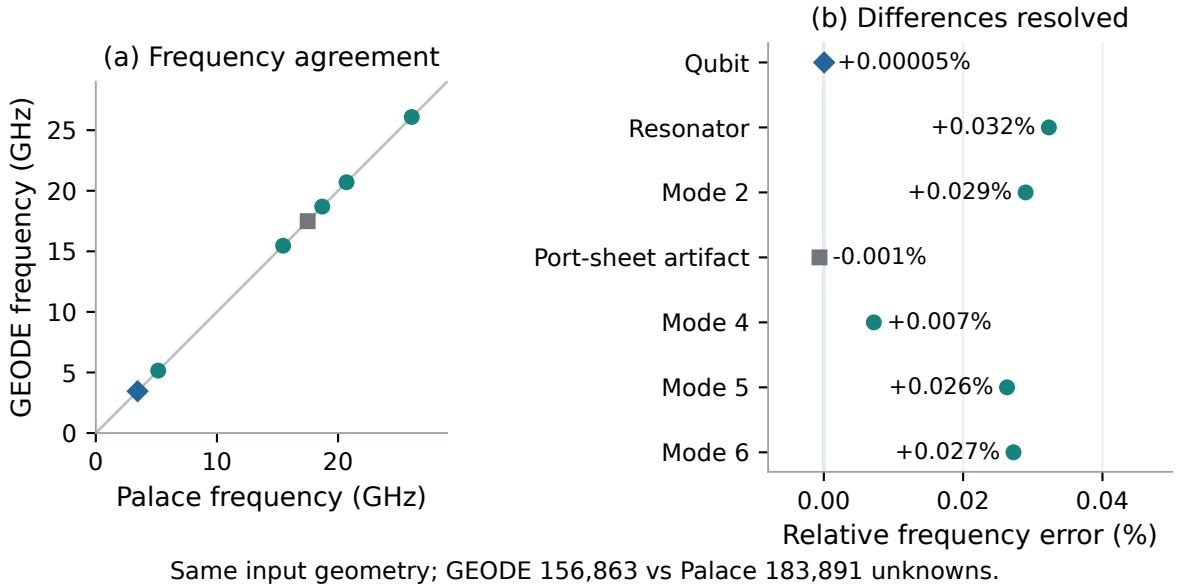


Figure 2: Frequency agreement from the same input geometry, including the qubit recovered by Palace with a 2 GHz search target. The six other modes use the original 4.5 GHz-target reference. (a) Frequency pairs and the identity line. (b) Signed relative errors, $100(f_{\mathrm{GEODE}}/f_{\mathrm{Palace}} - 1)$, computed from the committed frequencies. The 17.49 GHz port-sheet artifact is marked separately. Matching this artifact is not validation of its physics.

a relative 6.5×10^{-13}, so the local build does not move the eigenvalues. Table 2 gives the result (`benchmarks/transmon_eigen/results.toml`, `[oracles.palace_target2]`; raw outputs under `reference/fixtures/transmon_palace/results_p1_target2/`).

Palace finds the mode at 3.452749 GHz with port EPR 0.99424; geode-fem's committed value is 3.452751 GHz with $p = 0.9942$. The two frequencies agree to 5×10^{-5} %. Under L doubling the mode moves to 2.444899 GHz in Palace, a ratio of 0.70810, against geode-fem's committed 0.7081. Palace's value lies 0.280% below the lumped estimate at L and 0.140% below it at $2L$. That estimate takes C_Σ from the electrostatic branch on the same coarse mesh, so it is a consistency check between this paper's eigenmode and LOM branches, not an independent anchor. Every other mode in the re-run matches the committed Palace spectrum, and two further modes converge above it (33.384 and 33.896 GHz).

Earlier versions explained the supposed absence by saying that the lumped-shunt model gives the junction only its own 5.5 fF, so that its resonance lands at 17.5 GHz, and that the pad capacitance of a physical transmon (Koch et al., 2007) was missing. That explanation was wrong. The full-wave eigenproblem contains the island's capacitance through the field itself, which is why a mode exists in which the junction inductance resonates against the whole island.

What this does not show. This is not the converged physical qubit frequency. Both solvers start from the same coarse 133k-tetrahedron mesh at first order, and they do not solve the same discrete problem: Palace refines around the metal sheets and solves 183,891 unknowns against geode-fem's 156,863 (Section 4.1). The mesh is also far from converged for the capacitance that sets this frequency: one uniform refinement lowers the scalar-ε C_Σ by 26.3% (Section 7). The lumped estimate goes as $1/\sqrt{C_\Sigma + C_J}$, so we infer that the qubit frequency is not converged on this mesh

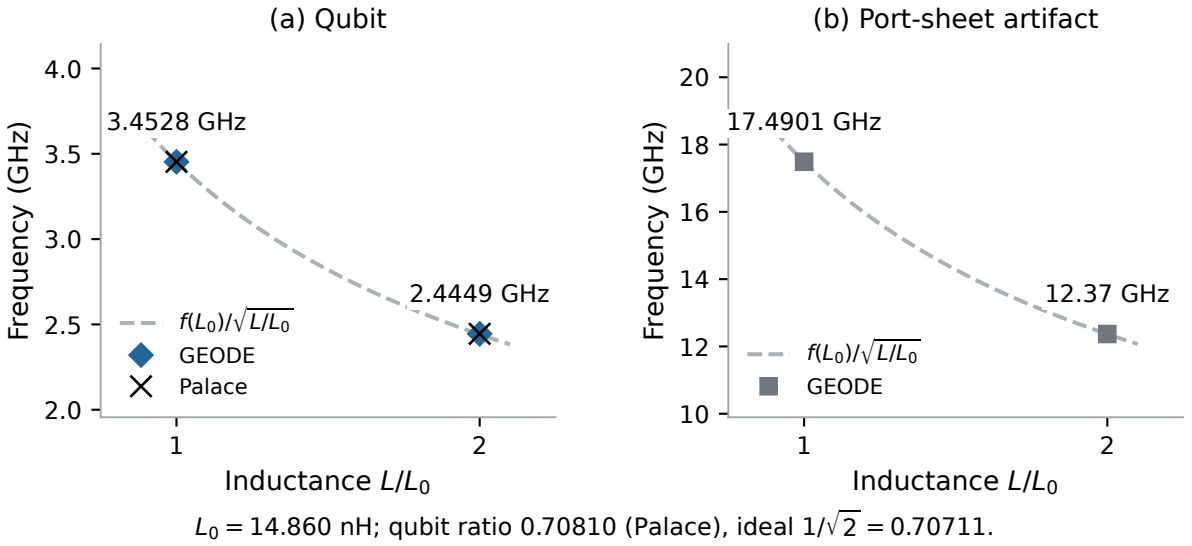


Figure 3: Inductance-doubling checks for (a) the qubit and (b) the port-sheet artifact, with $L_0 = 14.860\,\text{nH}$. Dashed curves show $f(L_0)/\sqrt{L/L_0}$, anchored independently at each baseline. Quoted GEODE tripwire values retain their stored precision. Palace's qubit frequencies, 3.452749 and 2.444899 GHz, give a ratio of 0.70810; the markers overlap GEODE's values at this scale. Both branches respond to L, so this check alone cannot identify a physical qubit mode.

either; that inference has not been tested by a refined eigensolve. The DeviceLayout.jl blog workflow (Peairs and Carson, 2025) reports a spectrum from 4.14 to 5.591 GHz that includes a ∼4 GHz qubit mode. We do not claim to reproduce it, and we do not attribute the difference from our 3.45 GHz to any single cause.

The committed artifacts still carry the old labels. `benchmarks/transmon_eigen/results.toml` keeps this mode under the section name `[spurious_mode]` with `excluded = true`, and the CPU timing artifacts label it `spurious_port`. The re-baselining of those classifications is tracked separately (issue #1045) and has not been done. We quote the artifact keys verbatim and state here what the evidence now shows.

6.2 The gauge and projection arc, re-read

Earlier versions told the following arc as the diagnosis of a spurious mode. Its measurements stand, every step committed (Section 12); what they mean changes once the two modes are correctly identified. geode-fem's headline Lanczos operates on the raw pencil of Eq. 2 without a gauge projection. (1) Classical tree–cotree elimination (Albanese and Rubinacci, 1988; Manges and Cendes, 1995) is not spectrum-preserving for the generalized eigenproblem — the resonator shifts 1.64% (5.1528 → 5.2372 GHz), outside our bar, pinned as a regression test. (2) The spectrum-preserving bulk divergence-free projector (structurally Palace's confinement) preserves the cavity spectrum (0.029%) and collapses the 13,747-mode gradient cluster. It removes the 17.49 GHz mode, which is almost entirely a gradient field (measured divergence ratio 50.2; projected norm 1.06×10^{-4}), and it keeps the 3.45 GHz mode, which is solenoidal with respect to the interior gradients (divergence ratio $\approx 6 \times 10^{-15}$). Earlier versions described this as the projector deflating the physical junction mode while the spurious mode survived. With the identities corrected it reads the other way: the projector

Table 2: The qubit mode in both solvers, from the identical input mesh at first order. Palace values are from the `Target` = 2 GHz re-run; geode-fem values are the committed ones that earlier versions reported as a spurious mode. The lumped estimate is $1/(2\pi\sqrt{L(C_\Sigma + C_J)})$ with the electrostatic $C_\Sigma = 136.684731$ fF of Section 7 and $C_J = 5.5$ fF. Palace's column is its port EPR; geode-fem's is the stiffness participation of Eq. 3. This is a same-input-mesh comparison on a coarse mesh, not a converged qubit frequency.

Quantity	Palace	geode-fem	Lumped estimate
Qubit mode, $L = 14.860$ nH (GHz)	3.452749	3.452751	3.46246
Port EPR / participation p	0.99424	0.9942	—
Qubit mode, $2L$ (GHz)	2.444899	2.4449	2.44833
L-doubling ratio ($1/\sqrt{2} = 0.70711$)	0.70810	0.7081	0.70711
17.49 GHz mode: port EPR / p	2.49×10^{-8}	1.000	—
17.49 GHz mode at $2L$ (GHz)	12.3674	12.37	—

kept the qubit mode and removed the port-sheet mode. (3) A port-aware rank-1 re-admission of one gradient direction restores the 17.49 GHz mode and returns all six of Palace's `Target` = 4.5 GHz modes at $\leq 0.029\%$ worst case. That construction did what it was built to do. Its purpose, though, was to retain a mode that Section 6.3 identifies as an artifact, because the Palace reference then in hand contained that mode and did not contain the qubit mode.

The remaining committed characterization of the 3.45 GHz mode is unchanged and now reads as a description of the qubit mode: under the L-doubling harness it moves $3.4528 \to 2.4449$ GHz (ratio 0.7081) with 99.4% of its stiffness energy in the K_{port} term. Earlier versions added that Palace's `LumpedPort` represents the port differently and therefore never sees this mode. That was also wrong: Palace's port operator has the same structure (Section 4.2), and Palace sees the mode.

The headline comparison deliberately remains the committed ungauged benchmark. The subsequent port-subspace implementation removes the near-zero survivor and admits the port-patch gradient direction with one shifted factorization. On this fixture, requesting seven modes at 4.5 GHz returns the six modes of Table 1 (worst relative frequency error 0.0289%) plus the 3.452751 GHz qubit mode; requesting six still misses the highest of the six. The committed local comparison records mean process CPU times of 46.32 s for the older eight-mode extraction route and 23.25 s for the new seven-mode route. These are geode-only measurements on a loaded Apple M3 Ultra, not a Palace comparison or a same-box update to the EC2 timings (Section 11.1). geode-fem's benchmark harness still classifies modes by frequency matching against the Palace reference; no reference-free rule is built. Figure 4 shows the spectrum.

6.3 The 17.49 GHz mode: a zero-voltage resonance of the distributed port term

Proven by committed data. Palace's port EPR for the 17.49 GHz mode is 2.49×10^{-8}, in the original run and in the re-run. After L doubling the mode moves to 12.3674 GHz (ratio 0.70711) and its port EPR is 2.94×10^{-8}. Palace builds that EPR from the port voltage, so in Palace this mode carries essentially no port voltage. In geode-fem the same mode has $p = 1.000$: all of its stiffness energy is in the port surface term. Both statements hold only if that energy sits in a surface pattern whose net voltage is zero.

Measured by a scratch probe. A probe on the committed fixture measured the uniform-voltage share of each mode's port energy in geode-fem: 0.00% for the 17.49 GHz mode and 100.00% for the 3.45 GHz mode. The probe source is committed (`docs/research/transmon-lumped-port-formulation.probe-1003.rs`) but it is scratch evidence: the build does not compile it, and no regression

test pins these two numbers.

Derived. The junction patch is four triangles with five nodes, of which only the centre node is not on a conductor. The gradient of that node's hat function is curl-free and has zero port voltage, but the isotropic surface term of Eq. 1 still stores energy in it. A field confined to the patch then has Rayleigh quotient $1/(LC)$, which is $f_{LC} = 17.60\,\mathrm{GHz}$, and the volume mass of the adjacent tetrahedra lowers it to 17.49 GHz. A two-terminal junction cannot carry this pattern. On this reading the mode is a sheet-internal resonance created by representing a lumped element as a distributed isotropic sheet, and both solvers have it because both use that representation. The reading predicts one such mode per free patch node, so a finer port mesh should show a cluster just below f_{LC}. That falsifier has not been run.

What it means for the credential. The 0.001% agreement on this mode is agreement on a shared artifact of the port model. It still tests that the two implementations assemble the same port operator, and nothing more. The physical content of the credential is the five cavity modes and the qubit mode.

Roadmap, not built. The planned remedy is a rank-1 port-voltage formulation (issue #1044): replace S_Γ in Eq. 1 by $b\,b^\mathsf{T}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\,dS$ the port-voltage functional, so that only the energy of the port voltage is kept. In the same scratch probe this formulation removes the 17.49 GHz mode (no mode remains between 15.47 and 18.69 GHz), leaves the 3.4528 GHz mode and the five cavity modes unchanged to four decimals, and makes the plain bulk projector exact (six modes from a six-mode request, residuals $\leq 1.7 \times 10^{-9}$, no reference filter). These are probe measurements on uncommitted solver code. No shipped solver path, committed benchmark or table in this paper uses the formulation. A solver using it would no longer reproduce Palace's 17.49 GHz mode, and whether it becomes the default is an open decision (issue #1046).

7 The Forward LOM Chain: From Fields to E_C

The lumped-oscillator-model (LOM) branch computes the transmon's Hamiltonian parameters from the *electrostatic* problem on the same committed mesh — and it is the chain the remainder of this paper differentiates. The mesh's metal is split into three node-disjoint conductors (ground+resonator, feedline, island); a tensor-ε electrostatic solve yields the Maxwell capacitance matrix by the energy-reaction form $C_{ij} = \phi_i^\mathsf{T} K \phi_j$; the island self-capacitance is reduced for the floating feedline ($C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$); and the charging energy follows as $E_C/h = e^2/(2C_\Sigma h)$, with the transmon spectrum evaluated charge-basis-exactly after Koch et al. (2007).

Measured on the committed fixture (`benchmarks/transmon_quantum/results.toml`): $C_\Sigma = 136.7\,\mathrm{fF}$, hence $E_C = 0.142\,\mathrm{GHz}$, $E_J/E_C = 77.6$ (with E_J from the junction inductance), $\omega_{01} = 3.38\,\mathrm{GHz}$, and anharmonicity $\alpha = -0.158\,\mathrm{GHz}$, with the $\alpha \approx -E_C$ sanity holding. The chain runs end-to-end from committed geometry. Its inputs are coarse-mesh values, as the next paragraph shows. The linear eigenproblem resolves the same oscillator at 3.45 GHz (Section 6.1); ω_{01} is its anharmonic $0 \to 1$ transition, computed from a C_Σ that omits the junction's own 5.5 fF, so the two numbers are not expected to coincide.

The committed mesh is not converged for C_Σ. One uniform refinement of the committed mesh splits every tetrahedron into eight (133,314 → 1,066,512). It lowers the scalar-ε C_Σ of the unmodified device from 137.706752 to 101.359397 fF, a drop of 26.3% (`benchmarks/transmon_diff opt/multiparam_results.toml`, `[mesh_refinement]`). The refined P1 space contains the coarse

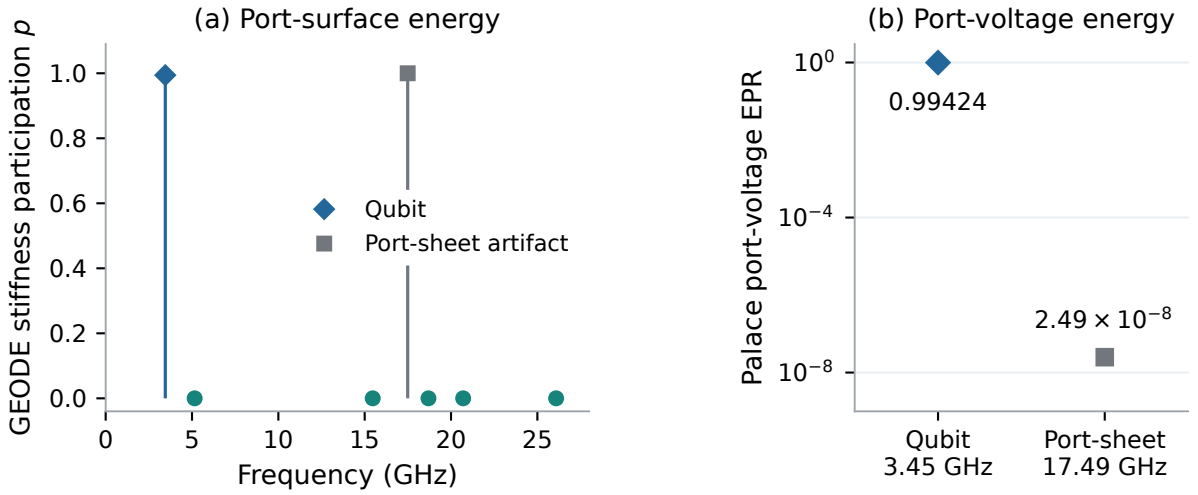


Figure 4: Two different port diagnostics on separate axes. (a) GEODE's stiffness participation counts all port-surface energy; both the qubit and the port-sheet artifact have participation near one. (b) Palace's port-voltage EPR is 0.99424 for the qubit but only 2.49×10^{-8} for the artifact. The right-hand axis is logarithmic. The two observables coincide only for uniform-voltage port fields; they must not be interpreted as interchangeable participation measures.

one under the same constraints, and C_Σ is an energy minimum, so refinement can only lower it. Each coarse value is therefore an *upper bound* on the converged value, and the shift of 36.34 fF is a lower bound on the coarse discretization error. Only one refinement level was run. The converged C_Σ is not measured, and the refined value does not bound it from below. A convergence study is in progress (issue #1051); this draft uses none of its results.

The refinement was measured for the scalar-ε model, the one the shape gradient differentiates. The tensor value of 136.7 fF above has not been re-solved on the refined mesh. The monotonicity argument applies to it equally, so it too is an upper bound; that its error is of similar size is an inference from the 0.75% coarse-mesh difference between the two models, not a measurement. It follows that $E_C = 0.142$ GHz is a coarse-mesh lower bound, and that E_J/E_C, ω_{01} and α will move when C_Σ converges. Every device-mesh C_Σ and E_C in Section 9 carries the same caveat.

The gap to the design anchor, and a withdrawn diagnosis. The extracted $C_\Sigma = 136.7$ fF *exceeds* the ∼90 fF design-anchor value back-solved from the blog workflow's expected $E_C = 0.2156$ GHz. Earlier versions, following the committed benchmark, diagnosed this as a model-definition difference and not a truncation artifact, on two checks: grounding the exterior wall moves C_Σ by a relative 6×10^{-5}, and the scalar-versus-tensor sapphire treatment shifts it 0.75%. Those two measurements stand, but neither tests mesh convergence, and we withdraw the diagnosis. The discretization error is at least 36.34 fF, the same order as the gap itself, and after one refinement the scalar-ε C_Σ of the unmodified device sits 12.7% above the 89.9 fF anchor instead of 53% above it. How much of the gap is discretization and how much is model definition cannot be said until the converged value is measured. We report the extracted numbers and the discrepancy; we do not retrofit to the anchor. Section 9 uses the anchor as an optimization target on the coarse mesh and reports what that does and does not establish.

Table 3: The merged 2×2 discrete-adjoint sensitivity matrix. Each cell: one forward + one adjoint solve sharing a single factorization; element Jacobians exact (forward-mode duals through the production kernels); validated against central finite differences of the full pipeline, with mutation tests. "FD rel-err" is the observed agreement of the analytic gradient with the FD reference in the merged test suite (test acceptance bounds are looser; see the module sources). Modules in `crates/geode-core/src/`.

Operator	Parameter	Module (issue / PR)	FD rel-err
scalar electrostatic (SPD)	material $\partial/\partial\varepsilon$	`adjoint` (#570 / #573)	$\sim 3 \times 10^{-8}$
scalar electrostatic (P1)	geometry $\partial/\partial X$	`shape` (#571 / #575)	$\sim 1 \times 10^{-9}$
driven $H(\mathrm{curl})$	material $\partial/\partial\varepsilon$	`driven::adjoint` (#576 / #579)	2.3×10^{-5} (worst region)
driven $H(\mathrm{curl})$	geometry $\partial/\partial X$	`driven::shape` (#577 / #581)	2.2×10^{-9}, 1.2×10^{-8} (two maps)

8 Differentiable Sensitivities: The Discrete-Adjoint Layer

8.1 The 2×2 sensitivity matrix

For a solve $A(\varepsilon, X)\, x = b(X)$ and a scalar observable $g(x)$, the discrete adjoint $A^{\mathsf{T}}\lambda = \partial g/\partial x$ turns the parameter gradient into explicit contractions:

$$\frac{dg}{d\varepsilon_k} = -\lambda^{\mathsf{T}} \frac{\partial A}{\partial \varepsilon_k} x, \qquad \frac{dg}{dX} = \lambda^{\mathsf{T}} \frac{\partial b}{\partial X} - \lambda^{\mathsf{T}} \frac{\partial A}{\partial X} x \tag{4}$$

(real case; the complex driven case takes $2\,\mathrm{Re}[\cdot]$ of the same contractions with the *transpose*, not conjugate-transpose, solve — the pencil is complex-symmetric, so the adjoint solve reuses the forward LU factorization). Everything on the right-hand side is assembly-level and therefore differentiable on the tensor substrate: the element Jacobians $\partial A/\partial X$ and $\partial b/\partial X$ are *exact*, computed by forward-mode dual numbers through the same closed-form element kernels the solver assembles — no finite-difference truncation enters the gradient. Each gradient costs one forward plus one adjoint solve sharing a single factorization (asserted programmatically: `n_factorizations` $= 1$).

Four adjoint modules are merged and validated, spanning material and geometry sensitivities on both operators (Table 3). Each carries a committed test that the analytic gradient match a central finite difference *of the full independent pipeline* (perturb the parameter, re-assemble, re-solve through the public solver path, recompute g) — and, load-bearingly, mutation tests proving the tolerances bite: the classic complex-adjoint conjugation error (A^{H} for A^{T}) is rejected by the FD check, as is a dropped term. The driven-shape case surfaced a physics subtlety worth recording: the current-source right-hand side $b = \mathrm{i}\omega\mu_0 \int N \cdot J$ is geometry-dependent, so the shape gradient carries a $\partial b/\partial X$ term absent from the material case — omitting it fails the finite-difference cross-check outright.

The matrix is deliberately 2×2: material and geometry are the two design-parameter classes, and the scalar and driven operators are the electrostatic (LOM) and RF (S-parameter/EPR-workhorse) branches of the solver. The LOM optimization of Section 9 exercises the scalar column end-to-end; the driven column is merged and validated at the operator level and composes with the driven solve that geode-fem's GPU path accelerates — RF shape optimization is the natural follow-on, not exercised here.

8.2 The capacitance$\to E_C$ chain, and a vanishing adjoint

Composing the shape adjoint with the LOM chain of Section 7 gives $\partial(C, E_C, \alpha)/\partial(\text{geometry})$ (merged as `shape::capacitance_shape_gradient`, #583 / PR #586). The composition carries an elegant simplification. The capacitance observable is the stored field energy, $C = 2W = \phi^{\mathsf{T}} K(X) \phi$, and W is *variationally stationary* in the solved potential ϕ: at the forward solution the free rows of

the cotangent $K\phi$ vanish (equilibrium), so the adjoint solution $\lambda \approx 0$ and the entire shape derivative collapses to the explicit-geometry term

$$\frac{dC}{dX} \;=\; \phi^{\mathsf{T}}\,\frac{\partial K}{\partial X}\,\phi \qquad \text{(at the solution)}, \tag{5}$$

a Hellmann–Feynman-like result: no adjoint back-solve is needed for the capacitance itself. The implementation keeps the general adjoint form (the mutation test measures the implicit part at $< 10^{-8}$ of the total, confirming the collapse numerically rather than assuming it), and chains through $\partial E_C/\partial C_\Sigma = -e^2/(2C_\Sigma^2 h)$ exactly. Validation is double: against the analytic parallel-plate law ($\partial C/\partial d = -\varepsilon_0\varepsilon_r$ per unit area under a gap scaling; $\partial C/\partial A = +2\varepsilon_0\varepsilon_r$ under an area scaling — opposite signs, both matched) and against central finite differences of the full pipeline, with observed relative errors at the 10^{-9} level.

9 Gradient-Based Optimization: Demonstrations and an Honest Negative

The initial demonstrations are a closed-loop proof on an analytic fixture, a single-parameter run on the real device mesh, and a three-parameter bounded optimizer on that mesh. Section 9.4 then tests optimization on a refined P2 discretization. The first two use damped Newton on $E_C(\theta) - E_C^{\text{target}}$ (equivalently $C(\theta)$) with the analytic $dE_C/d\theta$ from Section 8.2; each Newton step costs one forward plus one adjoint solve on a single factorization.

One caveat governs the initial coarse-mesh experiments below. Their capacitances are computed on the coarse committed mesh or on morphed copies of it, and that mesh overstates C_Σ: one uniform refinement lowers it by 26.3% (Section 7). The gradient validations and convergence statements below are exact statements about the coarse discrete problem. The finite-difference checks test the derivative of the discrete C_Σ, and the fresh solves test self-consistency on one discretization. None of them is evidence about the converged device, and the distances to the 89.9 fF anchor are distances measured in coarse-mesh capacitance. The framing claim is capability and step-count, not wall clock: a derivative-free loop (the parameter sweeps of current practice) spends N_{params} extra forward solves per step to estimate what the adjoint delivers exactly for one. No timing comparison against HFSS/Palace-based optimizers is measured or claimed.

9.1 Loop proof: Newton to a target E_C on a parallel-plate fixture

On a parallel-plate capacitor fixture ($\varepsilon_r = 3$; 343 nodes, 1,296 tets; θ scales the gap, so $d = L(1+\theta)$), Newton iteration drives the design from $C_0 = 120.0\,\text{fF}$ ($E_C = 0.1614\,\text{GHz}$) to the design-anchor target $E_C = 0.2156\,\text{GHz}$. Full Newton converges in *one* step — and we state the reason rather than dress it up: $C \propto 1/(1+\theta)$ makes $E_C(\theta)$ *affine* in θ, so one-step convergence is expected, and this tier proves the loop end-to-end (gradient, chain rule, update, convergence test), not a hard optimization. The converged $\theta = 0.335657909$ matches the closed-form optimum to 3×10^{-15}. The honesty check that matters: a *fresh, independent* forward solve at the converged geometry — assemble, solve, extract, no adjoint machinery — reproduces the target E_C to a relative error of 1.4×10^{-15}. A damped run ($\alpha = 0.5$) converges in 13 steps and supplies the descent illustration (Figure 7a). At the hit target, the Koch-exact spectrum gives $E_J/E_C = 51.0$, $\omega_{01} = 4.128\,\text{GHz}$, $\alpha = -0.247\,\text{GHz}$ — the $\sim$4 GHz qubit regime the design anchor implies. All trajectory values are committed in `benchmarks/transmon_diffopt/results.toml`.

9.2 The real device: validated gradients and deformation limits

The second tier runs on the committed 133k-tet SingleTransmon mesh itself, with a fixed-topology shape parameter θ that scales the island conductor in-plane about its centroid (101 island nodes

move; the mesh connectivity is fixed). Three results, in descending order of comfort (`benchmarks/transmon_diffopt/pad_results.toml`):

The gradient is real on the real mesh. At $\theta = 0$ the adjoint gradient $\partial C_\Sigma/\partial\theta = 198.198\,\mathrm{fF}$ is validated against a central finite difference of the *full independent pipeline* (move the island nodes, re-assemble, re-solve, re-extract): relative error 1.15×10^{-4} at $h = 10^{-4}$, with a clean $O(h^2)$ truncation sweep ($1.2 \times 10^{-2} \to 1.0 \times 10^{-3} \to 1.15 \times 10^{-4} \to 1.0 \times 10^{-5}$ as h falls $10^{-3} \to 3 \times 10^{-5}$). This is the paper's load-bearing real-geometry validation: the same machinery that passed at 10^{-9}-level on analytic fixtures holds up on production geometry.

The loop genuinely converges on the real mesh. A demonstration target within the mesh's validity budget ($C_\Sigma = 137.0\,\mathrm{fF}$) converges in two Newton steps — genuinely non-affine now (the gradient changes by ∼21% along the path) — and the converged design is confirmed by a fresh, independent multi-conductor extraction at relative error 5.6×10^{-6} (Figure 7b). The entire real-device run (validation sweep plus both optimization runs) takes 10.7 s.

The honest negative: the anchor is out of this parametrization's reach. The natural target is the ∼90 fF anchor of Section 7 (89.9 fF; $E_C = 0.2155\,\mathrm{GHz}$ — the last digit differs from the 0.2156 GHz quoted elsewhere because this run targets the rounded 89.9 fF rather than the back-solved 89.843 fF; both values are artifact-faithful). The $\theta = 0$ gradient puts it at $\theta \approx -0.241$ — but the fixed-topology map inverts its first tetrahedron at $\theta = -0.0097$, and the run's conservative budget (minimum tet-volume ratio 0.25) caps $\theta_{\mathrm{safe}} = -0.0073$, where the bounded Newton run stalls at $C_\Sigma = 136.5\,\mathrm{fF}$: a 33× shortfall on the coarse mesh. The diagnosis is geometric, not a gradient failure: the island's junction-attachment nodes sit ∼225 μm from the island centroid, so the centroid scale drags them against ∼0.7 μm junction-region tetrahedra. The 33× figure is a local linear estimate for the rigid island-only map, not a measured path to the anchor.

Harmonic motion extends the budget, but does not reach the anchor. Extending the prescribed island velocity into the volume with a P1 Laplace solve spreads the deformation; the other conductors and far boundary have zero velocity. The field D is computed once, giving $X(\theta) = X_0 + \theta D$. At the same minimum volume-ratio threshold 0.25, the safe negative step grows from −0.007258 to −0.064609, an 8.90× extension. The harmonic field takes 0.34 s to solve; the validation and bounded optimization run takes 7.8 s. The initial derivative is 209.087 fF per unit θ, with a full-pipeline central-FD relative error 1.054×10^{-4} at $h = 10^{-4}$. Bounded Newton reaches 128.792998 fF, independently confirmed by a fresh extraction, and stalls at the safe boundary, leaving 38.892998 fF to the anchor.

The initial-slope extrapolation in the artifact gives $\theta_{\mathrm{anchor}} = -0.2286$, or 3.54× the harmonic budget. That estimate understates the shortfall because the capacitance sensitivity falls along the measured path. From $dC/d\theta = -(C/E_C)\,dE_C/d\theta$, the two recorded steps give 209.1 and 114.5 fF per unit θ. *If the sensitivity does not recover beyond the measured endpoint*, the remaining gap needs at least another $38.893/114.5 \simeq 0.340$ in $|\theta|$, or a total approximately 0.404: at least 6.25× the safe budget (computed 6.256959, floored). This is a conditional extrapolation, not a measured bound on an uncomputed continuation. It is also a statement about the coarse mesh, whose C_Σ starts 47.8 fF above the anchor; once refined, the unmodified device starts about 11.5 fF above it (Section 7), so the 33× and 6.25× factors do not describe the converged problem. Both runs use scalar trace-averaged sapphire; neither establishes the tensor-material shape chain.

9.3 Three junction-pinned parameters: a methods result

The single-parameter runs stall because one scaling drags the junction leads against sub-micron tetrahedra. The third tier removes that constraint and adds a bounded optimizer. Its outcome is a methods result. It is not a device design, for reasons stated at the end of this subsection.

Gradients (proven on the coarse mesh). Three harmonic morph fields change the island length (θ_L), the pad width (θ_W) and the cutout gap (θ_G). Each is exactly zero on the junction nodes and on the island leads, so no parameter moves the junction region. The gradient of $C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$ with respect to all three comes from one assembly and one factorization. At $\theta = 0$ it is 93.427139, 44.855961 and -61.367367 fF per unit parameter, and the full-pipeline central finite difference at $h = 10^{-4}$ agrees to relative errors of 7.9×10^{-7}, 3.4×10^{-10} and 1.1×10^{-8} (`benchmarks/transmon_diffopt/multiparam_gradient.toml`). With the fields held fixed, moving along the joint corner of the three safe budgets does not reach the anchor: a fresh extraction at the largest safe fraction gives 97.027749 fF.

Bounded optimizer with per-step re-morphing (proven on the coarse mesh). The optimizer rebuilds the three morph fields about the current geometry at every accepted step. Each step is a projected minimum-norm Gauss–Newton step on the E_C residual under box, trust-region and linearized volume-ratio constraints. Every accepted step keeps the minimum tetrahedron volume ratio against the original mesh at or above 0.25 and is confirmed by an independent fresh extraction. Six morph variants were run in a fixed order (`benchmarks/transmon_diffopt/multiparam_results.toml`):

- fixed fields stall at 96.191952 fF, and unit re-morphing stalls at 94.053101 fF;
- volume-stiffened re-morphing with exponent $\alpha = 1$ and $\alpha = 2$ stalls at 92.063327 and 90.819818 fF;
- $\alpha = 3$ converges in 18 steps to 89.900001 fF, with an E_C residual of -3.019 Hz against a 10 kHz tolerance, a minimum volume ratio of 0.250635, fresh-solve agreement of 1.39×10^{-14}, and finite-difference agreement at the final design of 1.2×10^{-8}, 1.2×10^{-9} and 9.2×10^{-9};
- $\alpha = 4$ converges in 14 steps to 89.900413 fF.

The quantity that reaches 89.9 fF is the coarse-mesh scalar-ε C_Σ. At that design the tensor-ε value is 90.342806 fF; a separate outer retarget brings the tensor value to 89.899807 fF in two rounds, again on the coarse mesh.

What this result is not. *The design is not unique.* The $\alpha = 3$ and $\alpha = 4$ runs both reach the anchor with different geometries: island length 564.62 against 580.55 μm, pad width 8.68 against 7.84 μm, and cutout gap 67.65 against 67.83 μm (from 625.5, 24.0 and 30.0 μm). The mesh-quality floor is a property of the morph and not of the device, so the design reached depends on α.

The exponent was selected post hoc. The headline rule, the smallest swept α that converges, and the list of exponents were fixed after exploratory runs. All six runs are committed.

It is not a device-design result. On the once-refined mesh the $\alpha = 3$ design has $C_\Sigma = 66.609133$ fF, which is 25.9% *below* the anchor, while the unmodified device, refined, is 12.7% above it. The refinement shift at the headline design is 5,582 times the anchor tolerance. The 10 kHz convergence and the 10^{-14} fresh-solve agreement therefore measure the optimizer's self-consistency on one

discretization, not accuracy. The final dimensions are not an 89.9 fF device. The design has also not been regenerated by the layout toolchain or remeshed.

What the result does establish is the method: junction-pinned multi-parameter gradients that survive finite-difference checks on production geometry, a re-morphing scheme that extends the reachable range, and a bounded optimizer that converges on the discretization it is given.

Pending. **[PENDING issue-1051]** The mesh-converged C_Σ of this device, and with it the size of the true gap to the anchor, are being measured. The local P2 checks in Section 9.4 do not settle continuum convergence; no anchor-reaching claim is made for a certifiably converged discretization. **[PENDING issue-1037]** A second design target, the island–claw coupling ratio $\beta = C_g/C_\Sigma$, has not been run. The optimizer interface accepts several targets and is unit-tested with two targets and three parameters, but no two-target experiment on the device exists.

9.4 Refined P2 optimization and fixed-design validation

A refined extension optimizes the scalar-permittivity electrostatic capacitance with a neutral floating feedline, excluding the junction's 5.5 fF self-capacitance. Ten P1 flux-jump refinement cycles prepare 448,279 tetrahedra. P2 potentials are then solved with AMG-PCG in native f64 arithmetic. The three junction-pinned harmonic geometry fields are fixed and exactly prolonged between meshes; refinement is outside the differentiable geometry map. By stationarity, the P2 shape derivative contracts the equilibrated field with $\varepsilon[(\nabla\cdot v)|\nabla\phi|^2 - 2\nabla\phi^{\mathsf{T}}(\nabla v)\nabla\phi]$, integrated exactly on each affine tetrahedron.

Three accepted gradient steps change C_Σ from 78.387918 to 89.900518 fF (Figure 5). The minimum element-volume ratio over accepted steps is 0.61413. Fully re-solved central differences check all three final derivatives; the worst relative error is 3.87×10^{-7}. With the final geometry fixed, one extra local refinement to 528,324 tetrahedra gives 89.662755 fF, a 0.2645% shift; the three sensitivities change by at most 0.605%. Independent Palace P2 gives 89.900294 fF, differing by 0.000249%. Palace adds 3,777 interior-boundary bisections, so this comparison matches the physical polygonal geometry and coefficients, not the exact discrete mesh.

These results meet the working 1% local capacitance/sensitivity check. They do not certify continuum accuracy, tensor-permittivity accuracy, CAD regeneration, or full-wave frequency optimization. The P2/order-refinement choice followed exploratory P1 runs. A Palace P3 attempt was killed before producing a full capacitance matrix. Two Palace adaptive P2 steps primarily changed feedline capacitance and are not counted as a strong extra island-convergence check. The full optimization repeated with identical numerical results and exported mesh hash. Source-commit provenance, negative attempts and the final geometry are retained with the figure data; the frozen experimental record is linked in Appendix A.

10 Roadmap: Differentiating the Eigenmode Branch

The LOM branch above differentiates an *electrostatic* solve. The transmon's dynamical figures-of-merit — resonator frequency, participation-ratio and EPR self/cross-Kerr (Minev et al., 2021a; Nigg et al., 2012) — live in the *eigenmode* problem, and that branch is **not differentiable yet** in geode-fem. The paper commits to the explicit split:

Branch	Observables	Differentiable?
LOM (electrostatic)	E_C, $\alpha \approx -E_C$, coupling C	**yes** (Sections 8–9)
eigenmode / EPR	resonator f, participation, Kerr	**not yet** (this section)

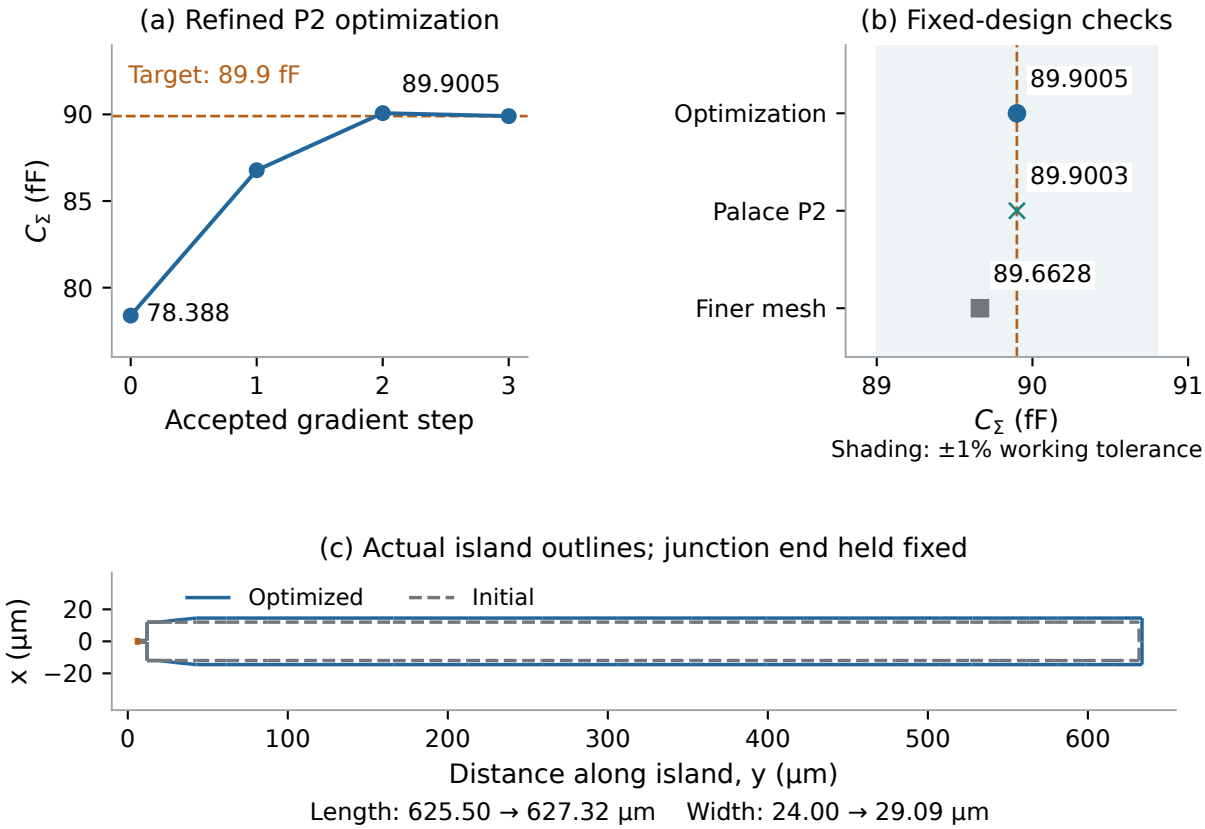


Figure 5: Refined electrostatic inverse design. (a) Three accepted gradient steps approach the 89.9 fF target on the P2 optimization mesh. (b) Fresh fixed-design checks using independent Palace P2 and a finer GEODE mesh; shading denotes the working ±1% tolerance, not a confidence interval or an error estimate. (c) Actual initial and optimized island outlines, rotated for display with equal axis scales; the junction end remains fixed. Length and width are measured from the exported polygonal geometry. These are local refinement and solver-agreement checks, not a certified continuum result.

Differentiating an eigenpair needs the adjoint-eigenpair (Hellmann–Feynman / Nelson) formulas (Nelson, 1976), which require solving the interior eigenproblem robustly at the physical shift — and geode-fem's matrix-free interior eigensolve is blocked at exactly that shift. The blocker is measured and committed (`benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md`): at $\sigma = 4.5$ GHz on the 133k-DOF fixture, a *single* inner AMS-preconditioned MINRES solve (outer Lanczos step 0) never converges — it plateaus on a nearly flat residual tail below $\sim 10^{-5}$ and was still running at 14,300 inner iterations ($\|r\| \approx 2.6 \times 10^{-6}$) when killed. The plateau is *coarse-solve-invariant* — an exact coarse solve stalls the same way — so the limiter is the SPD-proxy preconditioner $K + |\sigma|M$ itself, not the coarse solve. Nor is there a single inner tolerance that is both cheap and correct: at an inner tolerance of 10^{-2} the entire eigensolve completes in $\sim$56 s but returns the non-physical $\lambda \approx 0$ gradient near-kernel cluster (0.42–0.64 GHz), not the physical band. The named path — not built, cited as method, not result — is a preconditioned Helmholtz-projected Jacobi–Davidson iteration, applying the gradient-kernel projection inside the correction equation, for which Liang et al. (2026) provide the current adaptive-multilevel treatment of exactly this singular Maxwell eigenproblem; combined with the Nelson/Hellmann–Feynman eigenpair derivatives, that is

the route to differentiable eigenmode/EPR quantities. Until it lands, no claim in this paper extends to derivatives of the eigenmode spectrum.

11 Performance Context

The timings below qualify the cost of the capability. Eigenmode, driven CPU and driven GPU results are separate experiments; none measures the optimizer's cost or establishes a general solver ranking.

11.1 CPU eigenmodes: same box, verified mode content

The earlier timing request, six modes at 4.5 GHz, did not establish equivalent work. In the mode-logged re-run it returns four near-kernel modes, the 3.45 GHz qubit mode and only one of Palace's six modes in geode-fem, whereas Palace, which searches only above its target, returns its six. The historical run has no committed mode log or CPU-time record; its mode content and actual thread count cannot be recovered. We therefore withdraw the earlier claim that one geode core beats eight Palace ranks with roughly twelve times fewer core-seconds. This does not change the separate frequency agreement credential of Section 5.

The replacement experiment ran on one otherwise idle EC2 `m6i.4xlarge` in `us-east-1` on 2026-10-08, with geode `61e8571e` and Palace `fba6a5b`. One-wide runs were OS-pinned to one CPU; eight-wide runs used one hardware thread per physical core. Wall clock covers the whole solver process; core-seconds are measured user plus system CPU time, not wall time multiplied by requested width. Table 4 retains all three geode requests that recover the six reference modes. Two are tuned using the reference spectrum: requesting thirty modes at 4.5 GHz, or moving the shift to 20 GHz. The older port-aware extraction route does not use reference frequencies to choose its request, but returns two further modes alongside the six (one near-kernel mode and the qubit mode); matching the six still uses Palace. Palace's target is a lower bound, whereas geode's shift is a centre, so identical numerical requests need not select identical spectra.

Two asymmetries qualify every comparison in Table 4. First, the solvers do not solve the same number of unknowns. Palace refines the input mesh around the metal sheets, and every timed Palace run assembles 183,891 Nédélec unknowns against geode-fem's 156,863 (133,108 interior), about 17% more (Section 4.1). The table does not correct for this, and the difference favours geode-fem. Second, the reference set is Palace's six modes above 4.5 GHz. That set includes the 17.49 GHz port-sheet mode and excludes the qubit mode, which Palace cannot return at this target (Section 6). The "Extra modes" column counts returned modes outside that set. In the thirty-mode and port-aware rows one of the extras is the qubit mode, which the committed artifact still labels `spurious_port`; the others are near-kernel modes.

At eight wide Palace without field output takes 23.64 s, less than any of these geode routes (27.26–51.12 s). With field output, Palace takes 33.35 s and the two oracle-tuned routes are faster. At one wide the geode routes take less wall and CPU time, but neither comparison isolates a solver kernel: geode writes no fields and has no counterpart to Palace's error estimator, which remains enabled even with Save=0 (about 40.2 s at one rank and 5.0 s at eight). MPI busy-waiting also affects measured Palace core-seconds. Palace also carries about 17% more unknowns in every cell. No single route is selected as a headline winner.

The Palace build uses SuperLU-preconditioned GMRES, not AMS: the pinned source and build configuration resolve `Default` to SuperLU. A later, separate m6i session with a tuned Palace build found only a 1.01–1.05× same-box wall-time improvement over its untuned control; cross-session differences must not be credited to compiler tuning. Those runs also predate the new port-subspace solver. The local update in Section 6.2 cannot be substituted into this same-box timing table.

Table 4: Mode-verified CPU timings on one m6i.4xlarge: median wall seconds / measured core-seconds. All rows contain the six reference modes of Table 1. The solvers start from the same input mesh but Palace solves 183,891 unknowns and geode-fem 156,863; the timings are not normalized for that. Three repeats per cell except Palace one-rank Save=0 (one run). The two oracle-tuned requests are marked †; the historical port-aware route is the pre-port-subspace implementation. Palace always requests six modes above 4.5 GHz.

Request	Extra modes	1 wide	8 wide
geode 30 at 4.5 GHz†	24	34.45 / 34.44	29.08 / 40.71
geode 6 at 20 GHz†	0	33.21 / 33.20	27.26 / 38.87
geode port-aware 8 at 4.5 GHz	2	67.26 / 67.25	51.12 / 91.72
Palace, Save=6	0	186.84 / 186.82	33.35 / 266.66
Palace, Save=0	0	113.00 / 112.98	23.64 / 188.89

Table 5: Driven CPU-f64 linear-solve seconds on one r6i.4xlarge, one core/rank, median of two geode and three Palace runs, except 897,650 edges (one run each). geode includes operator assembly, preconditioner setup, iteration and residual check; Palace sums its Setup, Preconditioner and Linear Solve timers, excluding Operator Construction. These are not whole-process timings.

Edges	geode Jacobi	geode AMS	Palace AMS	Palace SuperLU
25,695	2.4	0.7	1.0	1.4
102,024	22.2	5.3	5.0	8.9
462,520	227.3	56.4	26.4	85.9
897,650	624.3	181.3	53.3	235.8

At 1,157,564 DOFs, a historical geode direct run completed on a 128 GB box in 565.5 s with 92.2 GB peak RSS. Palace's earlier eight-rank run recorded 423.12 s and about 33 GB aggregate memory. These are different-host observations, and the geode request lacks mode-content verification. The DOF count is geode-fem's; we infer that Palace's internal refinement applies to that mesh as well, and its unknown count for that run is not quoted here. They establish the direct factorization's large memory requirement, not a controlled performance crossover or equivalent physical-mode throughput.

11.2 CPU driven solve: AMS at scale

A separate f64 experiment uses assembled COCG on the lossy driven cube at $\omega = 0.10$, on one dedicated `r6i.4xlarge`, with geode `64d44269`. AMS reduces one-core linear-solve time relative to Jacobi by 3.4–4.2× across 25,695–897,650 edges. It reduces iterations from 2969–13662 to 39–105, but does not eliminate iteration growth. Every geode solve meets an explicitly recomputed relative-residual tolerance of 10^{-8}.

At 462,520 edges AMS needs about 3.4 GB versus Jacobi's 1.05 GB. At 897,650 edges it needs 8.5 GB versus 2.0 GB, and its setup alone takes 53.488 s. Eight threads reduce setup but do not accelerate geode's Krylov iteration; at 462,520 edges its AMS solve takes 50.4 s versus Palace AMS's 4.2 s on eight ranks. The one-core geode AMS advantage at the smallest size does not persist at scale: Palace AMS takes 53.3 s against 181.3 s at the largest size. These measurements concern one structured first-order fixture, assembled CPU operators and low frequencies. They establish neither GPU performance nor a remedy for the interior eigenproblem.

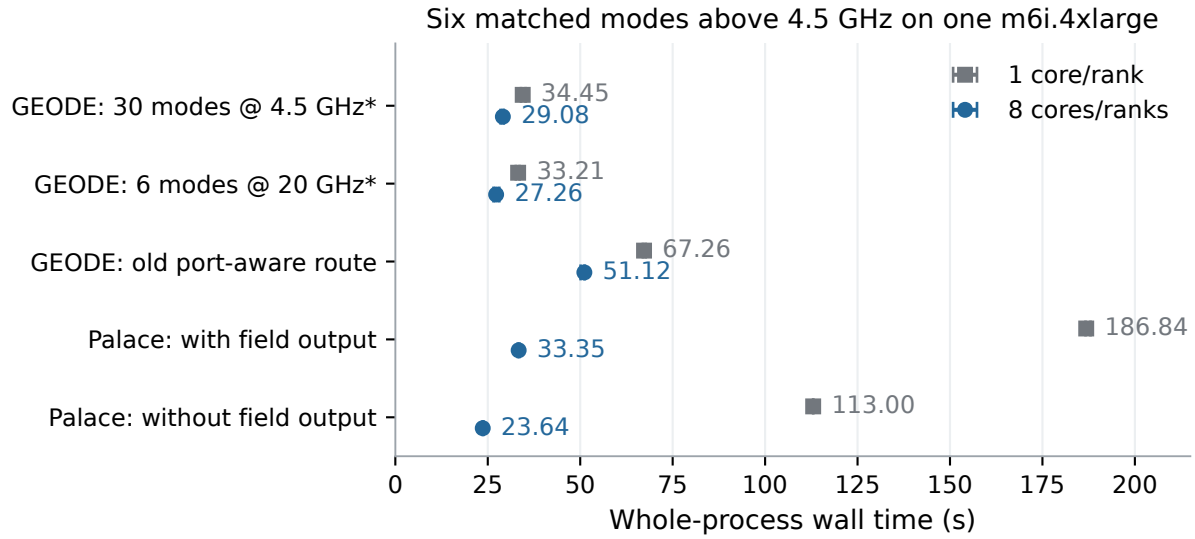


Figure 6: Whole-process wall time on one m6i.4xlarge, using a common linear axis for one and eight cores/ranks. Markers are medians; whiskers span the observed minimum and maximum (Table 4). All requests recover the six matched modes above 4.5 GHz, including the port-sheet artifact. Starred GEODE requests were tuned to the Palace spectrum. Palace carries about 17% more unknowns and retains error estimation even without field output; these are workload comparisons, not isolated solver-kernel speedups.

11.3 GPU cell: correctness proven, scaling an honest negative

geode-fem's GPU path accelerates the *driven* solve (matrix-free apply + on-device COCG), not the eigensolve of the credential, and all GPU execution is CUDA-f32 (Section 3). On 2026-07-14, geode-fem code executed on a physical GPU for the first time (EC2 `g6e.xlarge`, 1× NVIDIA L40S 46 GB, driver 595.71.05, CUDA 13.2): the two CUDA-f32 correctness smokes — the matrix-free Nédélec matvec and the on-device COCG — pass against the f64 CPU reference within f32-appropriate tolerances (15.0 and 3.3 s wall, durations dominated by GPU init and kernel JIT — correctness evidence, not performance).

The scaling study (committed as `benchmarks/gpu_driven_scaling/results.toml`; a σ-lossy lumped-port cube fixture at fixed frequency, mesh-refined from 1,854 to 25,695 edges, all sixteen cells on the same host) is an honest negative: **GPU-f32 matrix-free loses to every CPU configuration at every measured size** (Table 6). Computed from the artifact's unrounded medians, the assembled-CSR COCG on CPU is 136× faster at 1,854 edges (displayed: 0.032 versus 4.39 s) and still 44× faster at 25,695 (1.86 versus 81.8 s); even the direct LU is 13.5× faster at the top size (6.04 versus 81.8 s). The mechanism is kernel-launch domination (∼28 ms per iteration at the top size versus ∼0.63 ms for assembled-CSR on CPU) — 25.7k edges is far too small to fill an L40S — and f32 accuracy compounds the verdict (relative error degrading to 1.2×10^{-3} with refinement; nondeterministic convergence at the top size). One directional positive survives, and it is the one the architecture predicts: against the *same matrix-free algorithm on CPU*, the GPU's deficit falls monotonically to parity at ∼26k edges (81.8 versus 82.1 s). We state the GPU position as trajectory, not achievement, with tracked levers (kernel fusion; on-device f64/mixed-precision when cubecl ships it; preconditioning) — and we prefer publishing the bound to implying the win.

Table 6: Driven-solve wall-clock scaling, solve-only medians of $n = 3$ timed repetitions (warm-up excluded), same host (`g6e.xlarge`: 4 vCPU + 1× L40S). One fixture (σ-lossy lumped-port cube) at fixed ω, mesh-refined across the columns. f64 iterative configurations at tolerance 10^{-8}; the f32 GPU configuration at 10^{-6} (its recurrence floor sits above 10^{-8}). The accuracy row is the full-field relative L_2 error of the GPU-f32 solution against the Direct-f64 reference. At 25,695 edges the GPU-f32 configuration converged in the committed single-frequency runs but not in its five-point frequency sweep, and run-to-run convergence is nondeterministic at that size (f32 floor). Values verbatim from the committed `benchmarks/gpu_driven_scaling/results.toml`.

	Mesh size (Nédélec edges)			
Configuration	1,854	5,859	13,428	25,695
Direct faer LU (CPU f64), s	0.024	0.203	1.540	6.036
Assembled-CSR COCG (CPU f64), s	0.032	0.198	0.709	1.865
Matrix-free COCG (CPU f64), s	1.652	8.885	30.234	82.056
Matrix-free COCG (GPU f32), s	4.388	13.351	34.381	81.764
GPU-f32 accuracy vs. Direct-f64 (rel. L_2)	1.26×10^{-4}	2.93×10^{-4}	4.60×10^{-4}	1.20×10^{-3}

This cell is a cube fixture family, not the transmon eigenproblem, and its CPU baselines (4-vCPU host) are not comparable to the `m6i` numbers of Table 4.

12 Reproducibility

Every artifact behind every result in this paper is committed and scripted.[3] The exceptions are labelled where they occur: the scratch-probe measurements and source readings of Section 6 trace to a committed design note and probe source, not to a regenerable benchmark artifact.

- **Geometry and mesh.** Generated by DeviceLayout.jl v1.15.0's `SingleTransmon` example (AWS Center for Quantum Computing, 2025); the fixture generator is committed, and the MSH 4.1 fixture is pinned by SHA-256 (`5b3ff4c3...b33dd`; full hash and provenance in the committed `transmon_smoke.provenance.txt`).
- **The MSH conversion gotcha.** MFEM's MSH 4.1 reader rejects the DeviceLayout multi-entity-block file, so the Palace runs consume an MSH 2.2 conversion (`gmsh -save -format msh2`). The conversion is node-preserving (all 22,684 nodes unchanged, physical groups preserved) and physics-neutral: eigenvalues reproduce bit-for-bit across the conversion. Documented in the committed `palace_config.provenance.txt` because it is exactly the kind of silent pipeline step a same-mesh claim must disclose. A second such step went undisclosed in earlier versions: Palace refines the mesh it reads (4,913 elements added by local bisection, 6,014 duplicated vertices) and solves 183,891 unknowns, as its committed run logs record (`reference/fixtures/transmon_palace/results_p1/palace_run_v22.log`).
- **Credential artifacts.** The Palace configuration and provenance under `reference/fixtures/transmon_palace/`; geode-fem's benchmark definitions, the agreement numbers, the L-doubling tripwire (17.4901 → 12.37 GHz, committed ratio 0.7071), the qubit-mode measurements (3.4528 GHz, $p = 0.9942$; L-doubling ratio 0.7081; stored under the artifact's `[spurious_mode]` key), Palace's `Target` = 2 GHz re-run (`[oracles.palace_target2]`), and the full gauge/projection arc of Section 6.2 (tree–cotree counts and the 1.64% shift; bulk-projection divergence ratio 50.2 and projected norm 1.06×10^{-4}; port-aware acceptance

[3] Repository URL and archival DOI for the geode-fem source to be added at submission. TODO(operator).

at $\leq 0.029\%$) in `benchmarks/transmon_eigen/results.toml`, each pinned by a release-tier regression test. Palace's raw outputs (`eig.csv`, per-port participation, run log) are committed; a rerun reproduced `eig.csv` bit-for-bit. The re-run's configurations, logs, `eig.csv`, `port-EPR.csv` and provenance are under `reference/fixtures/transmon_palace/results_p1_target2/`. The source reading of Palace's port operator and search window, the port-voltage probe and the planned rank-1 formulation are in `docs/research/transmon-lumped-port-formulation.md`; its probe is scratch source that the build does not compile.

- **Precision of the committed agreement numbers.** `benchmarks/transmon_eigen/results.toml` stores geode-fem frequencies at three decimals while the Palace column carries full precision, so the committed per-mode `rel_err_pct` is computed against the rounded geode-fem values; at full precision the agreement is slightly better (approximately 0.029% worst case). We quote the committed-artifact numbers throughout and disclose the rounding here. Derived ratios quoted in the text (core-second and speedup multipliers) are computed from the displayed table values unless marked as committed.

- **LOM chain.** The three-conductor capacitance extraction and Koch-exact spectrum ($C_\Sigma = 136.7\,\text{fF}$, $E_C = 0.142\,\text{GHz}$, and the boundary and sapphire-model checks of Section 7) in `benchmarks/transmon_quantum/results.toml`; pinned by the release test `crates/geode-core/tests/transmon_quantum.rs`.

- **Adjoint layer.** The four sensitivity modules with their FD-and-mutation test suites: `crates/geode-core/src/adjoint.rs` (PR #573); `crates/geode-core/src/shape.rs` (PRs #575/#586; this module also carries `capacitance_shape_gradient`); `crates/geode-core/src/driven/adjoint.rs` (PR #579); and `crates/geode-core/src/driven/shape.rs` (PR #581).

- **Optimization demonstrations.** The parallel-plate loop proof (full trajectories, closed-form check, fresh-solve confirmation) in `benchmarks/transmon_diffopt/results.toml`; the real-device FD validation, demonstration convergence, and the bounded anchor attempt with its mesh-safety budget in `benchmarks/transmon_diffopt/pad_results.toml`; harmonic motion and its bounded anchor attempt in `benchmarks/transmon_diffopt/harmonic_results.toml`; the three-parameter gradients in `benchmarks/transmon_diffopt/multiparam_gradient.toml`; the bounded optimizer runs and the one-level mesh refinement in `benchmarks/transmon_diffopt/multiparam_results.toml`; all consumed by `crates/geode-core/tests/transmon_diffopt.rs`.

- **Performance artifacts.** The historical CPU cell in `benchmarks/transmon_bench_cpu/results.toml`; mode-verified timing in `benchmarks/transmon_bench_cpu/results_like_for_like_m6i.toml`; tuned-build controls in `benchmarks/transmon_bench_cpu/results_tuned_palace_m6i.toml`; local port-subspace evidence in `benchmarks/transmon_bench_cpu/results_like_for_like_local.toml`; driven CPU AMS results in `benchmarks/gpu_driven_scaling/results_ams_cpu_r6i.toml`; the 1.16M-DOF completion run log at `benchmarks/transmon_bench_cpu/geode_runs_1p16M_2026-07-15.log`; the GPU scaling cell (all sixteen cells, iteration counts, recomputed true residuals, DNF handling, same-host provenance) in `benchmarks/gpu_driven_scaling/results.toml`; the deep-shift eigensolve characterization at `benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md`.

- **Hardware disclosure.** CPU cell: EC2 `m6i.4xlarge` (8 physical cores / 16 vCPU, 64 GB), `us-east-1` for the mode-verified re-timing (the historical session was in `us-west-2`); driven

AMS: `r6i.4xlarge`, `us-east-1`; the 1.16M-DOF completion row: a 128 GB `r6i`-class box (see the committed log). GPU cell: EC2 `g6e.xlarge` (4 vCPU, 30 GB; 1× NVIDIA L40S 46 GB; driver 595.71.05, CUDA 13.2). The Palace `Target` = 2 GHz re-run: a local Apple M3 Ultra, Palace built from the in-repository Docker recipe (arm64), eight MPI ranks; no timing from it is used.

- **Versions.** geode-fem commit `3174015` (headline credential and historical CPU cells; the adjoint/diffopt layers are merged post-`3174015` and cited by PR number); Palace commit `fba6a5b`; DeviceLayout.jl v1.15.0; Burn/burn-cuda 0.21.

13 Discussion

What the credential does and does not establish. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997) the cross-validation is code verification and code-to-code comparison, not validation against nature: two independent implementations agreeing to $\leq 0.033\%$ from the identical input mesh, on discrete problems that differ by Palace's internal refinement, with scaling tripwires on the port operator and on the qubit mode. It does not certify that either solver matches a measured device — that is what Ye et al. (2025)-style experimental benchmarks address — but it rules out the large error class living in formulation, assembly, boundary handling, and eigensolver machinery. It establishes that geode-fem's eigenmode path reproduces a production solver from the same input. It does not establish mesh convergence. In particular the electrostatic observable that this paper differentiates is not converged on this mesh (Section 7), and the eigenmode credential says nothing about that.

What the differentiable-design evidence supports. The proven capability is bounded and we state its boundary. Proven: the 2×2 adjoint matrix at operator level (FD-validated with mutation tests); the capacitance$\to E_C$ chain with its stationarity collapse; a closed Newton loop confirmed by independent fresh solves on both an analytic fixture and the real 133k-tet mesh; a real-mesh gradient validated at 1.15×10^{-4} against the full independent pipeline; and, as a method, three junction-pinned gradients and a bounded re-morphing optimizer that converges on the coarse discretization (Section 9.3); and refined P2 optimization with local refinement and independent Palace checks (Section 9.4). Not proven, and not claimed: eigenmode/EPR derivatives (the Section 10 wall); a design that meets the anchor on a converged mesh (the coarse-mesh design sits 25.9% below it after one refinement, and the converged C_Σ is still being measured); two-target or coupling-constrained optimization; tensor-ε shape chains; multi-parameter optimization at scale; and any wall-clock superiority over sweep-based workflows. The capability is also not qubit-specific — shape and material sensitivities of electrostatic and driven-EM observables apply to RF and photonic device design generally; the transmon is the demonstration vehicle with the clearest documented unmet need.

Threats to validity. A common-mode error — both solvers wrong the same way — cannot be excluded by cross-comparison alone, and this paper now contains an example. The 17.49 GHz mode is an artifact of a port representation that both solvers share; the two agree on it to 0.001%, and earlier versions took that agreement, with the f_{LC} anchor, as confirmation of the junction physics. The same versions asserted the absence of a mode that Palace had not been asked for. Agreement with a reference bounds implementation error, not shared modeling choices or the reference's own search window. The early device-mesh capacitances are coarse-mesh upper bounds. The refined P2 checks provide local stability evidence, but neither experiment certifies the continuum device. The

three-parameter optimizer's stiffening exponent was chosen after exploratory runs, and its design is not unique. No two-target or coupling-constrained optimization is demonstrated. The parallel-plate tier's one-step convergence is an affine-map property, disclosed as such. The real-device chain uses the scalar trace-averaged sapphire (tensor-ε delta quantified at 0.75%). Repeat counts and timing boundaries differ across experiments and are stated in each table. The large-scale eigenmode completion used a different 128 GB box and lacks mode-content verification; it cannot establish equivalent-work throughput.

Limitations. (i) The eigenmode comparison covers one geometry and its coarse, first-order input mesh; no refined eigensolve has been run. The electrostatic branch now includes P2 and local refinement checks, but these do not establish continuum convergence. The initial P1 capacitance was far from converged. (ii) The lossless model leaves readout ports open; resistive terminations (hence complex-symmetric pencils and mode Qs) are out of scope. (iii) The credential eigenpath remains the ungauged committed benchmark, and mode classification still uses the Palace reference spectrum. The shipped port term retains the 17.49 GHz port-sheet artifact; the rank-1 port-voltage formulation that removes it in a probe is planned and not built. The committed artifacts still label the qubit mode as spurious. (iv) The CPU evidence depends on request selection, field output, error estimation and implementation version, and Palace solves about 17% more unknowns from the same input mesh; it supports no unconditional per-core or scaling advantage. (v) GPU evidence covers driven-solve correctness and an honest scaling negative in f32 on a cube fixture family; no GPU eigensolve path exists. (vi) The differentiable branch is electrostatic/LOM only (eigenmode/EPR is roadmap), currently scalar-ε for the shape chain, and its real-device demonstrations are bounded by the fixed-topology morph budget. The refined P2 result passes local checks but does not certify convergence of C_Σ.

14 Conclusion

The discrete-adjoint layer supplies solver-derived material and geometry gradients for electrostatic and driven FEM operators. Its stationary capacitance derivative closes the geometry-to-charging-energy loop, with fresh-solve confirmation on an analytic fixture and a within-budget real-device target. Harmonic motion extends the real-mesh safe deformation budget 8.90× without reaching the 89.9 fF anchor. A three-parameter junction-pinned optimizer with per-step re-morphing does reach it on the coarse mesh, as a methods result: that mesh overstates C_Σ by at least 36.34 fF, the design is not unique, and the converged value is still being measured. A refined P2 run reaches the target in three gradient steps and passes a held-out local refinement check with independent Palace agreement. This strengthens the numerical design evidence without certifying a continuum optimum. Same-input-mesh agreement with Palace supplies a forward-solver credential that now includes the transmon qubit mode, which earlier versions wrongly reported as absent. The updated mode-verified timing and driven AMS experiments delimit the computational cost without claiming a general performance advantage. Eigenmode/EPR derivatives remain future work.

Acknowledgments. geode-fem's development was AI-orchestrated (agent-driven implementation with human direction and review); this benchmark program and paper follow the same discipline, with every reported number traced to a committed, human-auditable artifact.

A Historical optimization examples and figure provenance

Figure 7 retains the analytic loop check and the earlier one-parameter device experiment. Its mesh-validity limit is specific to that geometry map and coarse discretization; it is not a limit on the refined optimization in Figure 5.

The refined figure uses a frozen snapshot of the recorded experiment and validation artifacts. Its co-located JSON data include source-commit and SHA-256 provenance; all other plots read the committed benchmark artifacts cited in their source scripts. Figures can be regenerated with Python, NumPy and Matplotlib, without a private plotting-style installation.

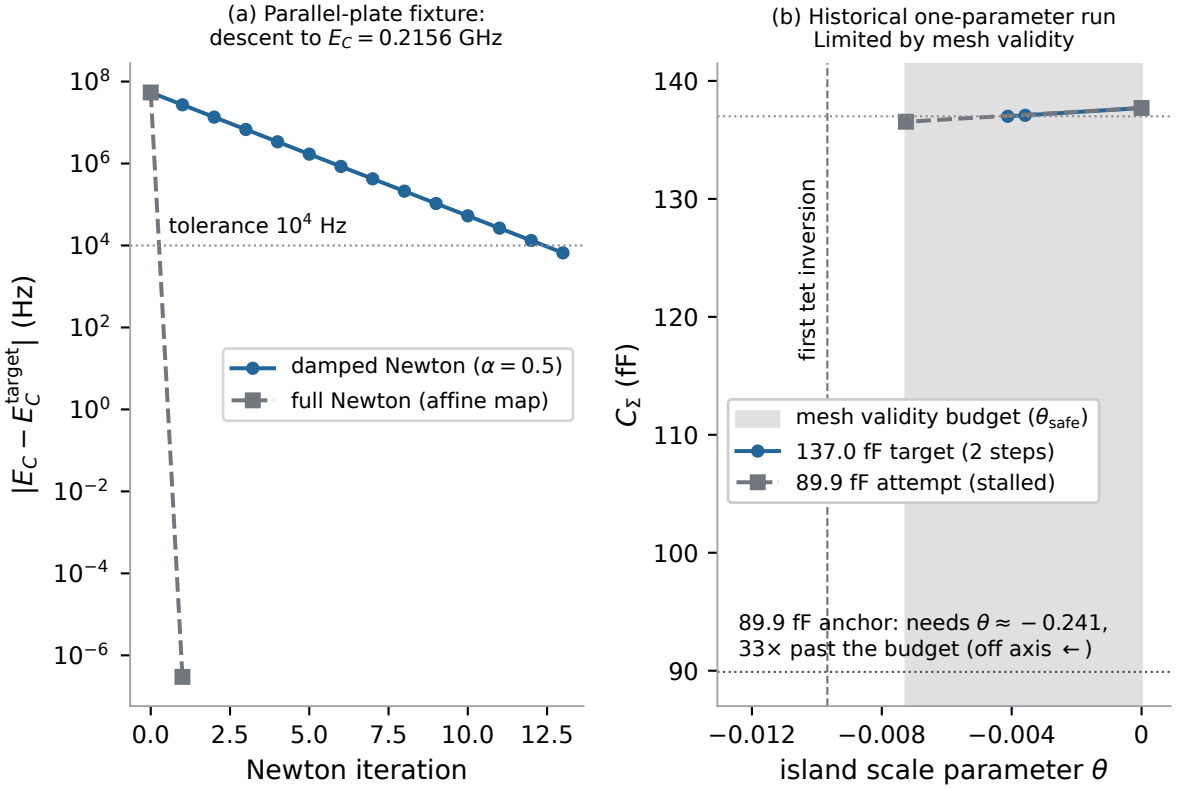


Figure 7: Gradient-based optimization to a target E_C via the discrete-adjoint shape gradient. (a) Parallel-plate loop proof: $|E_C - E_C^{\text{target}}|$ versus Newton iteration; full Newton converges in one step (the parametrization is affine — expected, and disclosed), the damped $\alpha = 0.5$ run supplies the 13-step descent curve; the converged design is confirmed by an independent fresh forward solve at relative error 1.4×10^{-15} (`benchmarks/transmon_diffopt/results.toml`). (b) The real 133k-tet DeviceLayout mesh under the fixed-topology island-scale parameter θ: a within-budget demonstration target ($C_\Sigma = 137.0\,\text{fF}$) converges in two genuine (non-affine) Newton steps, fresh-solve confirmed at 5.6×10^{-6}; the bounded run toward the 89.9 fF design anchor stalls honestly at the mesh validity budget $\theta_{\text{safe}} = -0.0073$ (first tetrahedron inversion at $\theta = -0.0097$), 33× short of the $\theta \approx -0.241$ the anchor requires (`benchmarks/transmon_diffopt/pad_results.toml`).

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